Advanced Topics in Computer Graphics II
Geometry Processing

Probabilistic Registration

January 7, 2025

Outline

Goal: find pairs of corresponding points between shapes represented as point sets to

- organize
- reconstruct
- synthesize

a broader class of shapes.

- **rigid** registration: find a map that preserves the distance between every pair of points
- **no-rigid** registration: find a map that does not preserve the distance between every pair of points but preserve the topology of the shapes

Probabilistic setting: Model uncertainty

- **Coherent point drift** (CPD) is a collection of different registration algorithms based on a Gaussian Mixture Model (GMM).
- performance, scalability to large data sets and stability/robustness of the correspondence estimation due to soft-matching

Probability Theory

Idea: Model uncertainty, which arises through

- noise on measurements
- and finite size of data sets

We model uncertainty by analyzing the occurrences of subsets of events from an event space of random experiments. We formulate random variables which describe certain outcomes of random experiments, which can take the values from the respective subsets of the event space.

Assigning a probability to the random variables leads to a probability distribution over the event space that can be interpreted as the relative likelihood that the value of the random variable would equal that value.

The probability that the value of a **random variable** x will lie in an interval (a, b) is given by

$$p(x \in (a,b)) = \int_a^b p(x)dx$$

Because probabilities are non-negative and the value of x must come from the event space, the **probability density** $p(x)$ must satisfy

$$p(x) \geq 0$$
$$\int_{-\infty}^{\infty} p(x)dx = 1$$

The **sum**- and **product-rule** in the discrete case

$$p(x) = \sum_y p(x,y)$$

$$p(x,y) = p(y|x)p(x)$$

take the following form in the continuous case

$$p(x) = \int p(x,y)dy$$

$$p(x,y) = p(y|x)p(x)$$

where

- $p(x)$ is called **marginal probability density**
- $p(x,y)$ is called **joint probability density**
- $p(y|x)$ is called **conditional probability density**

Given several continuous variables $x_1, \dots, x_D$, denoted collectively by the vector $\boldsymbol{x}$, we can define a **joint probability density** $p(\boldsymbol{x}) = p(x_1, \dots, x_D)$ such that the probability of $\boldsymbol{x}$ falling in an infinitesimal $\delta\boldsymbol{x}$ is given by $p(\boldsymbol{x})\delta\boldsymbol{x}$, satisfying

$$p(\boldsymbol{x}) \geq 0$$
$$\int_{-\infty}^{\infty} p(\boldsymbol{x}) d\boldsymbol{x} = 1$$

If the joint probability factorizes into the product of the marginals

$$p(x_1, \dots, x_D) = p(x_1) \cdot \ldots \cdot p(x_D)$$

then $x_1, \dots, x_D$ are called **independent**.

From the product rule and the symmetry property of $p(x, y) = p(y, x)$ we obtain a relationship between conditional probabilities called **Bayes' theorem**

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}$$

where the denominator can be expressed in terms of quantities appearing in the numerator

$$p(x) = \int p(x|y)p(y)dy$$

which is the normalization constant, required to ensure that $\int p(y|x)dx$

The average of some function $f(x)$ under a probability distribution $p(x)$ is called the **expectation** of $f(x)$

$$\mathbb{E}[f] = \int p(x)f(x)dx$$

Example: Given a finite number of points $|\{x_1, \ldots, x_n\}| = N$ and a uniform probability density $p(x) = \frac{1}{N}$ the expectation can be approximated as the mean

$$\mathbb{E}[f] \approx \frac{1}{N}\sum_{n=1}^{N} f(x_n)$$

The **variance** of a function $f(x)$ is defined as

$$\operatorname{var}[f] = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2] = \mathbb{E}[f(x)^2] - \mathbb{E}[f(x)]^2$$

and measures the variability around the mean value $\mathbb{E}[f(x)]$.

Given two variables x and y, the **covariance** is defined as

$$\operatorname{cov}[x, y] = \mathbb{E}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])] = \mathbb{E}[xy] - \mathbb{E}[x]\mathbb{E}[y]$$

For two vectors of random variables $\boldsymbol{x}$ and $\boldsymbol{y}$, the covariance is a matrix

$$\operatorname{cov}[\boldsymbol{x}, \boldsymbol{y}] = \mathbb{E}[(\boldsymbol{x} - \mathbb{E}[\boldsymbol{x}])(\boldsymbol{y} - \mathbb{E}[\boldsymbol{y}])^T] = \mathbb{E}[\boldsymbol{x}\boldsymbol{y}^T] - \mathbb{E}[\boldsymbol{x}]\mathbb{E}[\boldsymbol{y}]^T = \boldsymbol{\Sigma}$$

Given observed data $\boldsymbol{X}$ and model parameters $\boldsymbol{\theta}$

- capture assumptions about $\boldsymbol{\theta}$ before observing the data with **prior** $p(\boldsymbol{\theta})$
- observing the data $\boldsymbol{X} = \{\boldsymbol{x}_1, \ldots \boldsymbol{x}_N\}$ is expressed through the **likelihood** $p(\boldsymbol{X}|\boldsymbol{\theta})$
- evaluation of the uncertainty in $\boldsymbol{\theta}$ after observation gives the **posterior** $p(\boldsymbol{\theta}|\boldsymbol{X})$

Bayes' theorem takes the form

$$p(\boldsymbol{\theta}|\boldsymbol{X}) = \frac{p(\boldsymbol{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\boldsymbol{X})}$$

which can be stated as

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

where all of these quantities are viewed as functions of $\boldsymbol{\theta}$.

Note: The likelihood is not a probability distribution over $\boldsymbol{\theta}$, so the integral of the likelihood over $\boldsymbol{\theta}$ does not (necessarily) equal one.

Multivariate Normal distribution (Gaussian)

A **Multivariate Normal distribution (Gaussian)** of dimension D is defined as

$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D \det(\boldsymbol{\Sigma})}} \exp\Big(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\Big)$$

where the governing parameters are the mean $\boldsymbol{\mu}$ and the covariance matrix $\boldsymbol{\Sigma}$.Inhalt...

Suppose we have a data set $\boldsymbol{X} = \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_N\}$ representing N observations of value $\boldsymbol{x}$ i.i.d (**independently and identically distributed**) drawn from a Gaussian whose mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ are unknown.

We have seen that the joint probability of i.i.d variables is given by the product of the marginal probabilities and write the probability of the data set as

$$p(\boldsymbol{X}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \prod_{n=1}^{N} \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu},\boldsymbol{\Sigma})$$

We would like to determine the maximum likelihood

$$(\boldsymbol{\mu}^\star, \boldsymbol{\Sigma}^\star) = \arg\max_{\mu, \Sigma} p(\boldsymbol{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

for the mean $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ parameters from observations $\boldsymbol{X}$.
The **negative log likelihood-function** of $p(\boldsymbol{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$is given by

$$NLL = -\ln p(\boldsymbol{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{ND}{2}\ln(2\pi) + \frac{N}{2}\ln\det(\boldsymbol{\Sigma}) + \frac{1}{2}\sum_{n=1}^{N}(\boldsymbol{x}_n - \boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu})$$

Taking the derivative w.r.t the mean $\boldsymbol{\mu}$ we get

$$\frac{\partial}{\partial\boldsymbol{\mu}}NLL = \sum_{n=1}^{N}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \boldsymbol{\mu})$$

and setting to zero we obtain the maximum likelihood estimator

$$\boldsymbol{\mu}_{ML} = \frac{1}{N}\sum_{n=1}^{N}\boldsymbol{x}_n$$

Taking the derivative w.r.t the covariance $\boldsymbol{\Sigma}$ we have to take a short detour.[1]

The **derivative** of a scalar function $f(X)$, **with respect to the** $n \times m$-**matrix** X of independet variables, is given by

$$\frac{\partial f}{\partial \boldsymbol{X}} = \begin{pmatrix} \frac{\partial f}{\partial x_{11}} & \frac{\partial f}{\partial x_{12}} & \cdots & \frac{\partial f}{\partial x_{1m}} \\ \frac{\partial f}{\partial x_{21}} & \frac{\partial f}{\partial x_{22}} & \cdots & \frac{\partial f}{\partial x_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_{n1}} & \frac{\partial f}{\partial x_{n2}} & \cdots & \frac{\partial f}{\partial x_{nm}} \end{pmatrix}$$

For the directional derivative in the direction of matrix Y we get

$$\nabla_{\boldsymbol{Y}} f = tr\left(\frac{\partial f}{\partial \boldsymbol{X}} \boldsymbol{Y}\right)$$

Important examples of scalar functions of matrices include the trace of a matrix and the determinant.

[1] K. B. Petersen and M. S. Pedersen. *The Matrix Cookbook*. Version 20121115. Nov. 2012. URL: `http://www2.compute.dtu.dk/pubdb/pubs/3274-full.html`.

The **trace** of a square matrix $\boldsymbol{A}$ is defined to be the sum of the diagonal elements a_{ii} of $\boldsymbol{A}$

$$\mathrm{tr}[\boldsymbol{A}] = \sum_i a_{ii}$$

The trace is invariant under cyclical permutations of matrix products

$$\mathrm{tr}[\boldsymbol{ABC}] = \mathrm{tr}[\boldsymbol{CAB}] = \mathrm{tr}[\boldsymbol{BCA}]$$

where $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ are arbitrary matrices whose dimensions are conformal and are such that the product $\boldsymbol{ABC}$ is a square matrix.

The trace is very useful in calculating derivatives of **quadratic forms**, using the following "trick":

$$\boldsymbol{x}^T \boldsymbol{A}^{-1} \boldsymbol{x} = \mathrm{tr}[\boldsymbol{x}^T \boldsymbol{A}^{-1} \boldsymbol{x}] = \mathrm{tr}[\boldsymbol{x}\boldsymbol{x}^T \boldsymbol{A}^{-1}]$$

The derivative of $\mathrm{tr}[\boldsymbol{B}\boldsymbol{A}^{-1}]$ w.r.t $\boldsymbol{A}$ is is

$$\frac{\partial}{\partial \boldsymbol{A}} \mathrm{tr}[\boldsymbol{B}\boldsymbol{A}^{-1}] = -\boldsymbol{A}^{-T} \boldsymbol{B}^T \boldsymbol{A}^{-T}$$

For the quadratic form we get

$$\frac{\partial}{\partial \boldsymbol{A}} \boldsymbol{x}^T \boldsymbol{A}^{-1} \boldsymbol{x} = \frac{\partial}{\partial \boldsymbol{A}} \mathrm{tr}[\boldsymbol{x}\boldsymbol{x}^T \boldsymbol{A}^{-1}] = -\boldsymbol{A}^{-T} \boldsymbol{x}\boldsymbol{x}^T \boldsymbol{A}^{-T}$$

If $\boldsymbol{A}$ is symmetric we have

$$-\boldsymbol{A}^{-T} \boldsymbol{x}\boldsymbol{x}^T \boldsymbol{A}^{-T} = -\boldsymbol{A}^{-1} \boldsymbol{x}\boldsymbol{x}^T \boldsymbol{A}^{-1}$$

We also need to take the derivative of the logarithm of a determinant. In the following we will show that

$$\frac{\partial}{\partial \boldsymbol{A}} \ln \det(\boldsymbol{A}) = \boldsymbol{A}^{-T}$$

where $\boldsymbol{A}^{-T}$ is the inverse of $\boldsymbol{A}$ transposed.

Note that

$$\frac{\partial}{\partial a_{ij}} \ln \det(\boldsymbol{A}) = \frac{1}{\det(\boldsymbol{A})} \frac{\partial}{\partial a_{ij}} \det(\boldsymbol{A})$$

The determinant of a square matrix can be expanded using cofactors along the i-th row as

$$\det(\boldsymbol{A}) = \sum_j a_{ij} c_{ij},$$

where $\boldsymbol{C}$ is the matrix of **cofactors** with $c_{ij} = (-1)^{i+j} M_{ij}$ and M_{ij} is the minor associated with but independent of matrix element a_{ij}. Therefore, the derivative of the determinant $\det(\boldsymbol{A})$ w.r.t a_{ij} is just

$$\frac{\partial}{\partial a_{ij}} \det(\boldsymbol{A}) = c_{ij}$$

This matrix is simply the transpose of the matrix of cofactors and we have

$$\frac{\partial}{\partial a_{ij}} \ln \det(\boldsymbol{A}) = \frac{1}{\det(\boldsymbol{A})} \underbrace{\frac{\partial}{\partial a_{ij}} \det(\boldsymbol{A})}_{c_{ij}} = \frac{1}{\det(\boldsymbol{A})} c_{ij}$$

Recall that for any square matrix of oder n we have

$$\boldsymbol{A}\boldsymbol{C}^T = \det(\boldsymbol{A})\boldsymbol{I}_n$$

and therefore, if $det(A) \neq 0$ the formula for the matrix inverse is

$$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})}\boldsymbol{C}^T$$

and we get

$$\frac{\partial}{\partial \boldsymbol{A}} \ln \det(\boldsymbol{A}) = \frac{1}{\det(\boldsymbol{A})}\boldsymbol{C} = A^{-T}$$

Finally, for the maximum likelihood for the covariance matrix $\boldsymbol{\Sigma}$, ignoring irrelevant terms we get

$$\begin{aligned}
\frac{\partial}{\partial \boldsymbol{\Sigma}} NLL &= \frac{\partial}{\partial \boldsymbol{\Sigma}} \left(\frac{N}{2} \ln \det(\boldsymbol{\Sigma}) + \frac{1}{2} \sum_{n=1}^{N} (\boldsymbol{x}_n - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}_n - \boldsymbol{\mu}) \right) \\
&= \frac{\partial}{\partial \boldsymbol{\Sigma}} \left(\frac{N}{2} \ln \det(\boldsymbol{\Sigma}) + \frac{1}{2} \sum_{n=1}^{N} \mathrm{tr}[(\boldsymbol{x}_n - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}_n - \boldsymbol{\mu})] \right) \\
&= \frac{N}{2} \boldsymbol{\Sigma}^{-T} + \frac{\partial}{\partial \boldsymbol{\Sigma}} \left(\frac{1}{2} \sum_{n=1}^{N} \mathrm{tr}[(\boldsymbol{x}_n - \boldsymbol{\mu})(\boldsymbol{x}_n - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}] \right) \\
&= \frac{N}{2} \boldsymbol{\Sigma}^{-1} - \frac{1}{2} \boldsymbol{\Sigma}^{-1} \left(\sum_{n=1}^{N} (\boldsymbol{x}_n - \boldsymbol{\mu})(\boldsymbol{x}_n - \boldsymbol{\mu})^T \right) \boldsymbol{\Sigma}^{-1}
\end{aligned}$$

Setting to zero and multiplying by $\boldsymbol{\Sigma}$ from both sides yields the maximum likelihood estimator

$$\boldsymbol{\Sigma}_{ML} = \frac{1}{N} \sum_{n=1}^{N} (\boldsymbol{x}_n - \boldsymbol{\mu})(\boldsymbol{x}_n - \boldsymbol{\mu})^T$$

Maximum Likelihood for Gaussian Mixture Model

A mixture of Gaussians with K components is defined as

$$p(\boldsymbol{x}) = \sum_{k=1}^{K} \phi_k \underbrace{\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{C_k}$$

where the mixing coefficients $\phi_k \in [0, 1]$ are normalized:

$$\sum_{k=1}^{K} \phi_k = 1$$

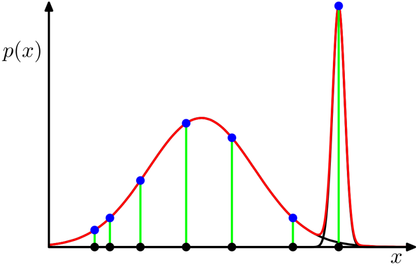


Figure: 1D example of GMM

For a set of observations $\boldsymbol{X} = \{\boldsymbol{x}_1, \ldots \boldsymbol{x}_N\}$ (i.i.d) the GMM will be the product

$$p(\boldsymbol{X}) = \prod_{n=1}^{N} p(\boldsymbol{x}_n) = \prod_{n=1}^{N} \left(\sum_{k=1}^{K} \phi_k \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right)$$

- The cost function is the negative log-likelihood (NLL) for the observation set $\boldsymbol{X}$

$$
\begin{aligned}
NLL &= -\ln p(\boldsymbol{X}) \\
&= -\ln\left(\prod_{n=1}^{N}\left(\sum_{k=1}^{K}\phi_k\mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)\right)\right) \\
&= -\sum_{n=1}^{N}\ln\left(\sum_{k=1}^{K}\phi_k\mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)\right)
\end{aligned}
$$

- Find parameters that minimize the cost, and maximize the likelihood by setting the derivative of the cost w.r.t. the different parameters to zero
- This is much more complex than for a single Gaussian because of the sum inside the logarithm.
- If we would happen to know the **posterior probabilities** $p(C_k|\boldsymbol{x}_n)$, which can be viewed as the responsibility that component k takes for 'explaining' the observation x and therefore are the responsibility that component k takes for 'explaining' the observation x and therefore are called **responsibilities** of the components, we could just use the previous approach for each component individually.

- Since we don't know the **responsibilities** we estimate them.
- Interpret the mixing coefficients as the prior probability of a random observation to come from the k^{th} component C_k among the K possible $p(C_k) = \phi_k$
- Each component $C_k = \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) = p(\boldsymbol{x}_n|k)$ corresponds to the likelihood function, describing the probability of observing x_n given component C_k
- The **responsibilities** and can be expressed using Bayes' theorem

$$p(C_k|\boldsymbol{x}_n) = \frac{p(C_k)p(\boldsymbol{x}_n|C_k)}{p(\boldsymbol{x}_n)} = \frac{\phi_k \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{l=1}^{K} \phi_l \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)} = p_{nk}$$

They model how much each component k is responsible for the result $\boldsymbol{x}_n$

- Derivative w.r.t. mean $\boldsymbol{\mu}_k$ given responsibilities p_{nk}

$$\frac{\partial(NLL)}{\partial\boldsymbol{\mu}_k} = \boldsymbol{\Sigma}_k^{-1}\sum_{n=1}^{N} p_{nk}(\boldsymbol{x}_n - \boldsymbol{\mu}_k)$$

- Equated to zero and reorganized the mean μ_k can be expressed as

$$\boldsymbol{\mu}_k = \frac{\sum_{n=1}^{N}\boldsymbol{x}_n p_{nk}}{N_k}$$

where

$$N_k = \sum_{n=1}^{N} p_{nk}$$

- Derivative w.r.t. covariance variance $\boldsymbol{\Sigma}_k$

$$\frac{\partial(NLL)}{\partial \boldsymbol{\Sigma}_k} = \frac{N_k}{2}\boldsymbol{\Sigma}_k^{-1} - \frac{1}{2}\boldsymbol{\Sigma}_k^{-1}\left(\sum_{n=1}^{N} p_{nk}(\boldsymbol{x}_n - \boldsymbol{\mu}_k)(\boldsymbol{x}_n - \boldsymbol{\mu}_k)^T\right)\boldsymbol{\Sigma}_k^{-1}$$

- Equated to zero and expressed with N_k the covariance $\boldsymbol{\Sigma}_k$ can be written as

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k}\sum_{n=1}^{N} p_{nk}(\boldsymbol{x}_n - \boldsymbol{\mu}_k)(\boldsymbol{x}_n - \boldsymbol{\mu}_k)^T$$

- Maximize NLL with respect to the mixing coefficients under the constraint $\sum_k \phi_k = 1$ using a Lagrangian multiplier λ

$$\frac{\partial}{\partial \phi_k} NLL = \frac{\partial}{\partial \phi_k}\left(-\sum_{n=1}^{N} \ln\left(\sum_{l=1}^{K} \phi_l \mathcal{N}(x_n|\mu_l \Sigma_l)\right)\right) = -\lambda \frac{\partial}{\partial \phi_k}\left(\sum_{l=1}^{K} \phi_l - 1\right)$$

computing the derivative and setting to zero gives

$$0 = \sum_{n=1}^{N} \frac{\mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{l=1}^{K} \phi_l \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)} + \lambda \qquad \forall k$$

$$\Leftrightarrow \phi_k \lambda = -\sum_{n=1}^{N} \frac{\phi_k \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{l=1}^{K} \phi_l \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)} \qquad \forall k$$

$$\Leftrightarrow \lambda \underbrace{\sum_{k=1}^{K} \phi_k}_{=1} = -\sum_{n=1}^{N} \frac{\sum_{k=1}^{K} \phi_k \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{l=1}^{K} \phi_l \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)} = -N$$

- Using $\lambda = -N$ we can solve for ϕ_k

$$
\begin{aligned}
0 &= \sum_{n=1}^{N} \frac{\mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{l=1}^{K} \phi_l \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)} - N \\
N &= \sum_{n=1}^{N} \frac{\mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{l=1}^{K} \phi_l \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)} \\
\phi_k N &= \sum_{n=1}^{N} \frac{\phi_k \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{l=1}^{K} \phi_l \mathcal{N}(\boldsymbol{x}_n|\boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)} \\
\phi_k N &= \sum_{n=1}^{N} p_{nk} = N_k \\
\phi_k &= \frac{N_k}{N}
\end{aligned}
$$

Maximum likelihood in the general case: Expectation Maximisation Algorithm (EM)

- Iterative method for maximum likelihood estimation (when there is no analytic solution)
- Alternate the expectation (E-)step and maximization (M-)step until convergence
- E-step: estimate expected responsibilities based on current model parameters
- M-step: estimate new parameters using current responsibilities
- E-step following M-step is guaranteed to increase the log-likelihood

Initialize mixing coefficients using a uniform distribution

$$\phi_k = \frac{1}{K}$$

and the covariances as the total unbiased covariance of the observations

$$\boldsymbol{\Sigma}_1 = \cdots = \boldsymbol{\Sigma}_K = \frac{1}{N-1} \sum_{n=1}^{N} (\boldsymbol{x}_n - \bar{\boldsymbol{x}})(\boldsymbol{x}_n - \bar{\boldsymbol{x}})^T$$

where $\bar{\boldsymbol{x}} = \frac{1}{N} \sum_{n=1}^{N} \boldsymbol{x}_n$.

Compute the responsibilities with the current mixing coefficients and variances

$$p_{nk}^{old} = \frac{\phi_k^{old} p^{old}(\boldsymbol{x}_n | C_k)}{p^{old}(\boldsymbol{x}_n)} = \frac{\phi_k^{old} \mathcal{N}(\boldsymbol{x}_n | \boldsymbol{\mu}_k^{old}, \boldsymbol{\Sigma}_k^{old})}{\sum_{l=1}^{K} \phi_l^{old} \mathcal{N}(\boldsymbol{x}_n | \boldsymbol{\mu}_l^{old}, \boldsymbol{\Sigma}_l^{old})}$$

building responsibility matrix $\boldsymbol{P}^{old}$ and set

$$N_k^{old} = \sum_{n=1}^{N} p_{nk}^{old}$$

In order to find the update rules for the parameters for the maximization step we look at the change of the cost function

$$
\begin{aligned}
NLL^{new} - NLL^{old} &= -\sum_{n=1}^{N} \ln p^{new}(\boldsymbol{x}_n) + \sum_{n=1}^{N} \ln p^{old}(\boldsymbol{x}_n) \\
&= -\sum_{n=1}^{N} \ln \frac{p^{new}(\boldsymbol{x}_n)}{p^{old}(\boldsymbol{x}_n)}
\end{aligned}
$$

Using Jensen's inequality

$$
\ln\left(\sum_k \lambda_k \boldsymbol{x}_k\right) \geq \sum_k \lambda_k \ln(\boldsymbol{x}_k)
$$

when $\lambda_k \geq 0$ and $\sum_k \lambda_k = 1$ we can see that this iteration scheme will find a minimum for the cost.

Introducing the factor $\frac{p_{nk}^{old}}{p_{nk}^{old}} = 1$ the responsibility p_{nk}^{old} in the nominator plays the role of λ_k in Jensen's inequality.

$$
\begin{aligned}
NLL^{new} - NLL^{old} &= -\sum_{n=1}^{N} \ln\left[\frac{\sum_{k=1}^{K} \phi_k^{new} p^{new}(\boldsymbol{x}_n|C_k)}{p^{old}(\boldsymbol{x}_n)} \frac{p_{nk}^{old}}{p_{nk}^{old}}\right] \\
&\leq -\sum_{n=1}^{N}\sum_{k=1}^{K} p_{nk}^{old} \ln\left[\frac{\phi_k^{new} p^{new}(\boldsymbol{x}_n|C_k)}{p^{old}(\boldsymbol{x}_n) p_{nk}^{old}}\right] =: \tilde{Q}
\end{aligned}
$$

It is evident that the new cost value $NLL^{new} \leq NLL^{old} + \tilde{Q}$, which means the cost will be minimized in each iteration until minimum.

Minimizing the cost is done by maximising the numerator of $\tilde{Q}$ denoted by

$$
Q = -\sum_{n=1}^{N}\sum_{k=1}^{K} p_{nk}^{old} \ln[\phi_k^{new} p^{new}(\boldsymbol{x}_n|C_k)]
$$

called the **objective function**.

Maximising the objective function Q by the same procedure as for the NLL, the following update rules for the parameters can be found

$$\boldsymbol{\mu}_k^{new} = \frac{1}{N_k} \sum_{n=1}^{N} \boldsymbol{x}_n p_{nk}^{old}$$

$$\boldsymbol{\Sigma}_k^{new} = \frac{1}{N_k} \sum_{n=1}^{N} p_{nk}^{old} (\boldsymbol{x}_n - \boldsymbol{\mu}_k)(\boldsymbol{x}_n - \boldsymbol{\mu}_k)^T$$

$$\phi_k^{new} = \frac{N_k^{old}}{N}$$

After E-step and M-step the new parameters override the old ones and the E- and M-steps are repeated until convergence (or a couple of times).

Convergence can be defined by a threshold under which the cost change should be before the algorithm is stopped

(Rigid, Affine) Coherent-Point-Drift

Let there be N points in X and M points in Y.

1. A target point $\boldsymbol{x}_n$ is either part of a target shape or an outlier with probability w
2. If $\boldsymbol{x}_n$ is an outlier it is generated from an outlier distribution $p_{out}(\boldsymbol{x}_n) = \frac{1}{N}$
3. If $\boldsymbol{x}_n$ is a non-outlier, an index $m \in \{1, \dots, M\}$, indicating that $\boldsymbol{x}_n$ corresponds to $\boldsymbol{y}_m$, is sampled with probability $p(m) = \frac{1}{M}$, where $\sum_m p(m) = 1$
4. Then $\boldsymbol{x}_n$ is generated from a D-dimensional multivariate normal distribution with mean $\mathcal{T}(y_m)$ and covariance matrix $\sigma^2 \boldsymbol{I}_D$

The target point set is generated by an N-times repetition of the steps $1 - 4$.

Define the GMM probability distribution for a point $\boldsymbol{x}$ as

$$p(\boldsymbol{x}) = \sum_{m=1}^{M+1} p(m)p(x|m) = wp_{out}(\boldsymbol{x}) + (1-w)\sum_{m=1}^{M} p(m)p(\boldsymbol{x}|m)$$

where $p(\boldsymbol{x}|m) = \mathcal{N}(\boldsymbol{x}|\boldsymbol{y}_m, \sigma^2)$ and $p(\boldsymbol{x}|M+1) := p_{out}(\boldsymbol{x}) = \frac{1}{N}$.

The GMM for the whole target point set is then

$$p(\boldsymbol{X}) = \prod_{n=1}^{N} \left(\sum_{m=1}^{M+1} p(m) p(\boldsymbol{x}_n | m) \right)$$

with negative log likelihood

$$NLL = -\sum_{n=1}^{N} \ln \sum_{m=1}^{M+1} p(m) p(\boldsymbol{x}_n | m)$$

We define the correspondence probability between $\boldsymbol{y}_m$ and $\boldsymbol{x}_n$ as the posterior probability of the GMM centroid given the data point as

$$p(m | \boldsymbol{x}_n) = \frac{p(m) p(\boldsymbol{x}_n | m)}{p(\boldsymbol{x}_n)}$$

which leads to the following objective function in the EM-framework

$$Q = -\sum_{n=1}^{N} \sum_{m=1}^{M+1} p^{old}(m | \boldsymbol{x}_n) \ln [p^{new}(m) p^{new}(\boldsymbol{x}_n | m)]$$

Rewriting the objective function in terms of transformation parameters $\boldsymbol{\theta}$ and σ^2 we get

$$Q(\boldsymbol{\theta}, \sigma^2) = \frac{1}{2\sigma^2} \sum_{n=1}^{N} \sum_{m=1}^{M} p^{old}(m|\boldsymbol{x}_n) \|\boldsymbol{x}_n - \mathcal{T}(\mathbf{Y}_m, \boldsymbol{\theta})\|^2 + \frac{N_P D}{2} \ln \sigma^2$$

with $N_P = \sum_m \sum_n p^{old}(m|\boldsymbol{x}_n) \leq N$ and

$$p^{old}(m|x_n) = \frac{\exp(-\frac{1}{2}\|\frac{\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{y}_m, \boldsymbol{\theta}^{old})}{\sigma^{old}}\|^2)}{\sum_{k=1}^{M} \exp(-\frac{1}{2}\|\frac{\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{y}_m, \boldsymbol{\theta}^{old})}{\sigma^{old}}\|^2) + c}$$

where $c = \sqrt{(2\pi\sigma^2)^D} \frac{w}{1-w} \frac{M}{N}$.

Minimizing Q with the EM-Algorithm necessarily decreases the NLL unless it is already at a local minimum.

Define the transformation of the GMM centroid locations as

$$\mathcal{T}(\boldsymbol{y}_m, \boldsymbol{R}, \boldsymbol{t}, s) = s\boldsymbol{R}\boldsymbol{y}_m + \boldsymbol{t}$$

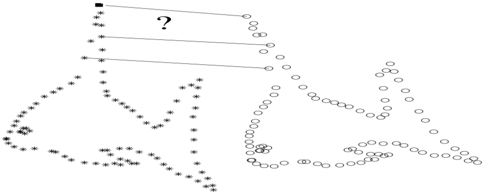
where $\boldsymbol{R} \in \mathbb{R}^{D\times D}$ is a rotation matrix, $\boldsymbol{t} \in \mathbb{R}^D$ a translation vector and s a scaling parameter.

The objective function takes the form

$$Q(\boldsymbol{R}, \boldsymbol{t}, s, \sigma^2) = \frac{1}{2\sigma^2} \sum_{m,n=1}^{M,N} p^{old}(m|\boldsymbol{x}_n) \|\boldsymbol{x}_n - s\boldsymbol{R}\boldsymbol{y}_m - \boldsymbol{t}\|^2 + \frac{N_P D}{2} \ln \sigma^2$$

$$\text{s.t. } \boldsymbol{R}^T\boldsymbol{R} = 1, \qquad \det(\boldsymbol{R}) = 1$$

The minimization of the objective function is difficult due to the constraints on $\boldsymbol{R}$.



To compute the optimal rotation matrix $\boldsymbol{R}$ aligning the two point sets under the current correspondences we first eliminate $\boldsymbol{t}$ from Q by taking the respective derivative and setting it to zero

$$\boldsymbol{t} = \frac{1}{N_P}\boldsymbol{X}^T\boldsymbol{P}^T\boldsymbol{1} - s\boldsymbol{R}\frac{1}{N_P}\boldsymbol{Y}^T\boldsymbol{P}\boldsymbol{1} = \boldsymbol{\mu}_x - s\boldsymbol{R}\boldsymbol{\mu}_y$$

where $\boldsymbol{P}$ has elements $p_{mn} = p^{old}(m|\boldsymbol{x}_n)$ and the mean vectors are defined as in matrix notation

$$\boldsymbol{\mu}_x = \mathbb{E}[\boldsymbol{X}] = \frac{1}{N}\boldsymbol{X}^T\boldsymbol{P}^T\boldsymbol{1}, \qquad \boldsymbol{\mu}_y = \mathbb{E}[\boldsymbol{Y}] = \frac{1}{N}\boldsymbol{Y}^T\boldsymbol{P}\boldsymbol{1}$$

Substituting $\boldsymbol{t}$ back into Q and writing it in matrix form gives

$$Q = \frac{1}{2\sigma^2}\left(\mathrm{tr}[\hat{\boldsymbol{X}}^T d(\boldsymbol{P}^T\boldsymbol{1})\hat{\boldsymbol{X}}] - 2s\cdot\mathrm{tr}[\hat{\boldsymbol{X}}^T\boldsymbol{P}^T\hat{\boldsymbol{Y}}\boldsymbol{R}^T]\right) + s^2\cdot\mathrm{tr}[\hat{\boldsymbol{Y}}^T d(\boldsymbol{P}\boldsymbol{1})\hat{\boldsymbol{Y}}] + \frac{N_P D}{2}\ln\sigma^2$$

where $\hat{\boldsymbol{X}} = \boldsymbol{X} - \boldsymbol{1}\boldsymbol{\mu}_x^T$ and $\hat{\boldsymbol{Y}} = \boldsymbol{Y} - \boldsymbol{1}\boldsymbol{\mu}_y^T$ are the centered point sets.

Using the **Singular Value Decomposition (SVD)** of the matrix

$$\boldsymbol{A} = \hat{\boldsymbol{X}}^T \boldsymbol{P}^T \hat{\boldsymbol{Y}} = \boldsymbol{U} \boldsymbol{\Lambda} \boldsymbol{V}^T$$

the optimal rotation[2][3][4] aligning $\boldsymbol{X}$ and $\boldsymbol{Y}$ is

$$\boldsymbol{R} = \boldsymbol{U} \boldsymbol{C} \boldsymbol{V}^T$$

where $\boldsymbol{C} = d(1, \ldots, 1, \det\left(\boldsymbol{U} \boldsymbol{V}^T\right))$ is a diagonal matrix.

[2] K. S. Arun et al. "Least-Squares Fitting of Two 3-D Point Sets". In: *IEEE Trans. Pattern Anal. Mach. Intell.* 9.5 (1987).

[3] S. Umeyama. "Least-Squares Estimation of Transformation Parameters Between Two Point Patterns". In: *IEEE Trans. Pattern Anal. Mach. Intell.* 13 (1991), pp. 376–380.

[4] O. Sorkine-Hornung and M. Rabinovich. "Least-squares rigid motion using svd". In: *Computing* 1.1 (2017), pp. 1–5.

Data: $\boldsymbol{X}$, $\boldsymbol{Y}$
Result: $\mathcal{T}(\boldsymbol{Y}) = s\boldsymbol{Y}\boldsymbol{R}^T + \mathbf{1}\boldsymbol{t}^T$
Initialization: $\boldsymbol{R} = \boldsymbol{I}$, $\boldsymbol{t} = 0$, $s = 1$, $0 \leq w \leq 1$;

$$\sigma^2 = \frac{1}{DNM} \sum_{n=1}^{N} \sum_{m=1}^{M} \|\boldsymbol{x}_n - \boldsymbol{y}_m\|^2;$$

while *not converged* **do**

E-step: Compute $\boldsymbol{P}$;

$$p_{mn} = \frac{\exp(-\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{Y})_m\|^2)}{\sum_{k=1}^{M} \exp(-\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{Y})_k\|^2) + (2\pi\sigma^2)^{D/2}\frac{w}{1-w}\frac{M}{N}};$$

M-step: Solve for $\boldsymbol{R}, s, \boldsymbol{t}, \sigma^2$;

$$N_P = \mathbf{1}^T\boldsymbol{P}, \boldsymbol{\mu}_X = \frac{1}{N_P}\boldsymbol{X}^T\boldsymbol{P}^T\mathbf{1}, \boldsymbol{\mu}_Y = \frac{1}{N_P}\boldsymbol{Y}^T\boldsymbol{P}\mathbf{1};$$

$$\hat{\boldsymbol{X}} = \boldsymbol{X} - \mathbf{1}\boldsymbol{\mu}_X^T, \hat{\boldsymbol{Y}} = \boldsymbol{Y} - \mathbf{1}\boldsymbol{\mu}_Y^T;$$

$$\boldsymbol{A} = \hat{\boldsymbol{X}}^T\boldsymbol{P}^T\hat{\boldsymbol{Y}}, \text{ compute SVD of } \boldsymbol{A} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{S}\boldsymbol{V}^T;$$

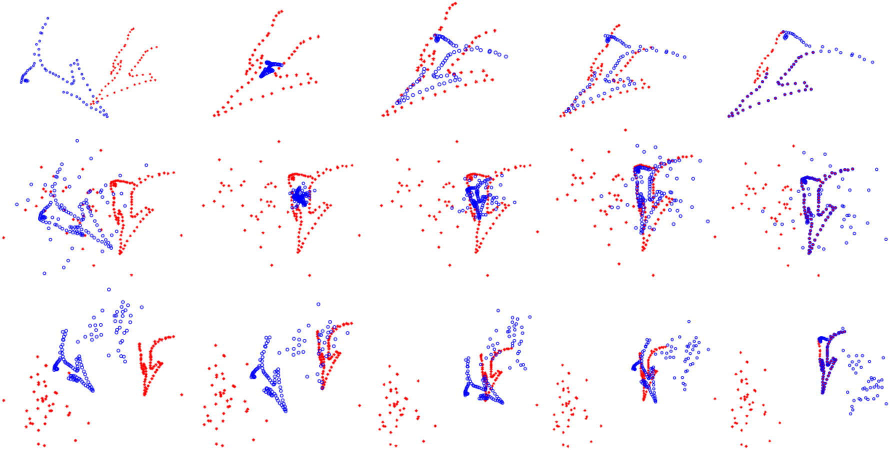
$$\boldsymbol{R} = \boldsymbol{U}\boldsymbol{C}\boldsymbol{V}^T, \text{ where } \boldsymbol{C} = d(1, \ldots, 1, \det\big(\boldsymbol{U}\boldsymbol{V}^T\big));$$

$$s = \frac{\mathrm{tr}[\boldsymbol{A}^T\boldsymbol{R}]}{\mathrm{tr}[\hat{\boldsymbol{Y}}^T d(\boldsymbol{P}\mathbf{1})\hat{\boldsymbol{Y}}]};$$

$$\boldsymbol{t} = \boldsymbol{\mu}_x - s\boldsymbol{R}\boldsymbol{\mu}_y;$$

$$\sigma^2 = \frac{1}{N_P D}(\mathrm{tr}[\hat{\boldsymbol{X}}^T d(\boldsymbol{P}\mathbf{1})\hat{\boldsymbol{X}}] - s \cdot \mathrm{tr}[\boldsymbol{A}^T\boldsymbol{R}]);$$

end



Initialization | Iteration 10 | Iteration 30 | Iteration 40 | Result (iteration 50)

Data: $\boldsymbol{X}$, $\boldsymbol{Y}$
Result: $\mathcal{T}(\boldsymbol{Y}) = \boldsymbol{Y}\boldsymbol{B}^T + \mathbf{1}\boldsymbol{t}^T$
Initialization: $\boldsymbol{B} = \boldsymbol{I}$, $\boldsymbol{t} = 0$, $0 \leq w \leq 1$;

$$\sigma^2 = \frac{1}{DNM}\sum_{n=1}^{N}\sum_{m=1}^{M}\|\boldsymbol{x}_n - \boldsymbol{y}_m\|^2;$$

while *not converged* **do**

E-step: Compute $\boldsymbol{P}$;

$$p_{mn} = \frac{\exp(-\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{Y})_m\|^2)}{\sum_{k=1}^{M}\exp(-\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{Y})_k\|^2) + (2\pi\sigma^2)^{D/2}\frac{w}{1-w}\frac{M}{N}};$$

M-step: Solve for $\boldsymbol{R}, s, \boldsymbol{t}, \sigma^2$;

$$N_P = \mathbf{1}^T\boldsymbol{P}, \boldsymbol{\mu}_X = \frac{1}{N_P}\boldsymbol{X}^T\boldsymbol{P}^T\mathbf{1}, \boldsymbol{\mu}_Y = \frac{1}{N_P}\boldsymbol{Y}^T\boldsymbol{P}\mathbf{1};$$

$$\hat{\boldsymbol{X}} = \boldsymbol{X} - \mathbf{1}\boldsymbol{\mu}_X^T, \hat{\boldsymbol{Y}} = \boldsymbol{Y} - \mathbf{1}\boldsymbol{\mu}_Y^T;$$

$$\boldsymbol{B} = (\hat{\boldsymbol{X}}^T\boldsymbol{P}^T\hat{\boldsymbol{Y}})(\hat{\boldsymbol{Y}}^T d(\boldsymbol{P}\mathbf{1})\hat{\boldsymbol{Y}})^{-1};$$

$$\boldsymbol{t} = \boldsymbol{\mu}_x - \boldsymbol{B}\boldsymbol{\mu}_y;$$

$$\sigma^2 = \frac{1}{N_P D}(\mathrm{tr}[\hat{\boldsymbol{X}}^T d(\boldsymbol{P}\mathbf{1})\hat{\boldsymbol{X}}] - \mathrm{tr}[\hat{\boldsymbol{X}}^T\boldsymbol{P}^T\hat{\boldsymbol{Y}}\boldsymbol{B}^T]);$$

end

Nonrigid Coherent Point Drift: Coherent Motion Theory

Idea: Points close to one another tend to move coherently; The displacement function v should be smooth.

Define the transformation as

$$\mathcal{T}(\boldsymbol{Y}, v) = \boldsymbol{Y} + v(\boldsymbol{Y})$$

Regularize this ill-posed problem using Tikhonov regularization. Adding a regularization term to the NLL, we obtain

$$f(v, \sigma^2) = NLL + \frac{\lambda}{2}\phi(v)$$

which comes in an Bayesian approach from a prior on displacement fields $p(v) = \exp(-\frac{\lambda}{2}\phi(v))$.

Using variational calculus the optimal functional form of v is a linear combination of particular kernel functions.

$$\boldsymbol{v} = \boldsymbol{G} \cdot \boldsymbol{W}$$

Using the L_2-norm as a regularizer on a displacement field in a Reproducing Kernel Hilbert Space (RKHS) with associated unique kernel G, the regularizer can be represented as

$$\|\boldsymbol{v}\|^2_{\mathbb{H}_m} = \int \frac{|\tilde{v}(\boldsymbol{s})|^2}{\tilde{G}(\boldsymbol{s})} d\boldsymbol{s}$$

due to the representer theorem. $\tilde{G}$ and $\tilde{v}$ are the Fourier transform of the kernel and the displacement function respectively.

This formulation of the norm in the Fourier domain is frequently used in Regularization theory.

- Smoothness of a function as a measure of oscillatory behavior
- Within the class of differentiable functions, one is smoother than another if it has less energy at high frequencies
- Measure high frequency content by first high-pass filtering the function and then measuring the resulting power
- $\tilde{G}$ represents a low-pass filter, which approaches zero as $\|\boldsymbol{s}\| \to \infty$

Writing the regularization term as

$$\phi(\boldsymbol{v}) = \|\boldsymbol{v}\|^2_{\mathbb{H}_m} = \|\mathcal{P}\boldsymbol{v}\|^2$$

using an operator $\mathcal{P}$ which "extracts" a part of the function for regularization.

This leads to the following objective function Q

$$Q(\boldsymbol{v}, \sigma^2) = \frac{1}{2\sigma^2} \sum_{n=1}^{N} \sum_{m=1}^{M} p^{old}(m|\boldsymbol{x}_n) \|\boldsymbol{x}_n - (\boldsymbol{y}_m + v(\boldsymbol{y}_m))\|^2 + \frac{N_P D}{2} \ln \sigma^2 + \frac{\lambda}{2} \|\mathcal{P}v\|^2$$

Ignoring all terms that are independent of v and writing Q in integral form, we get

$$Q(\boldsymbol{v}, \sigma^2) = \int \underbrace{\left(\frac{1}{2\sigma^2} \sum_{n,m}^{N,M} p^{old}(m|\boldsymbol{x}_n) \|\boldsymbol{x}_n - (\boldsymbol{y}_m + v(\boldsymbol{y}_m))\|^2 \delta(\boldsymbol{z} - \boldsymbol{y}_m) + \frac{\lambda}{2} v(\boldsymbol{z})^* \mathcal{P}^* \mathcal{P} v(\boldsymbol{z}) \right)}_{g(v(z),z)} d\boldsymbol{z}$$

A minimizing function $v(\boldsymbol{z})$ must satisfy the Euler-Lagrange differential Equation

$$\frac{\partial}{\partial v} g(v(\boldsymbol{z}), \boldsymbol{z}) = 0$$

We compute the partial derivative of g w.r.t v of the first term by writing the norm as a scalar product

$$\frac{\partial}{\partial \boldsymbol{v}} \frac{1}{2\sigma^2} \sum_{n,m}^{N,M} p^{old}(m|\boldsymbol{x}_n)[\boldsymbol{x}_n - (\boldsymbol{y}_m + v(\boldsymbol{y}_m))]^* [\boldsymbol{x}_n - (\boldsymbol{y}_m + v(\boldsymbol{y}_m))]\delta(\boldsymbol{z} - \boldsymbol{y}_m)$$

$$= -\frac{2}{2\sigma^2} \sum_{n,m}^{N,M} p^{old}(m|\boldsymbol{x}_n)(\boldsymbol{x}_n - (\boldsymbol{y}_m + v(\boldsymbol{y}_m)))\delta(\boldsymbol{z} - \boldsymbol{y}_m)$$

We compute the partial derivative of g w.r.t v of the second term by writing the norm as a scalar product

$$\frac{\partial}{\partial \boldsymbol{v}} \frac{\lambda}{2} [v(\boldsymbol{z})]^* [\mathcal{P}^* \mathcal{P} v(\boldsymbol{z})] = \frac{2\lambda}{2} \mathcal{P}^* \mathcal{P} v(\boldsymbol{z})$$

and we get the Euler-Lagrange Equation by summing up the two terms and equating to zero

$$\frac{1}{\sigma^2\lambda}\sum_{n=1}^{N}\sum_{m=1}^{M} p^{old}(m|\boldsymbol{x}_n)(\boldsymbol{x}_n - (\boldsymbol{y}_m + v(\boldsymbol{y}_m)))\delta(\boldsymbol{z} - \boldsymbol{y}_m) = \mathcal{P}^*\mathcal{P}v(\boldsymbol{z})$$

for all vectors $\boldsymbol{z}$, where $\mathcal{P}^*$ is the adjoint operator to $\mathcal{P}$.

Variational Solution

The solution to such a partial differential equation can be written as the integral transformation of its left side with a Green's function G of the self-adjoint operator $\mathcal{P}^*\mathcal{P}$ using $\mathcal{P}^*\mathcal{P} \cdot G(\boldsymbol{z}, \boldsymbol{y}_m) = \delta(\boldsymbol{z} - \boldsymbol{y}_m)$

$$v(\boldsymbol{z}) = \sum_{m=1}^{M} \underbrace{\frac{1}{\sigma^2 \lambda} \sum_{n=1}^{N} p^{old}(m|\boldsymbol{x}_n)(\boldsymbol{x}_n - (\boldsymbol{y}_m + v(\boldsymbol{y}_m)))}_{\boldsymbol{w}_m} G(\boldsymbol{z}, \boldsymbol{y}_m) = \sum_{m=1}^{M} \boldsymbol{w}_m G(\boldsymbol{z}, \boldsymbol{y}_m)$$

This solution is incomplete and in general we have

$$v(\boldsymbol{z}) = \sum_{m=1}^{M} \boldsymbol{w}_m G(\boldsymbol{z}, \boldsymbol{y}_m) + \psi(\boldsymbol{z})$$

where $\psi(\boldsymbol{z})$ lies in the null space of $\mathcal{P}$. Choosing Gaussians with precision β as kernel we have

- The Green's function is also a Gaussian
- The Kernel matrix is positive definite
- The null space term $\psi(\boldsymbol{z}) = 0$

Nonrigid-CPD

Data: $\boldsymbol{X}$, $\boldsymbol{Y}$
Result: $\mathcal{T}(\boldsymbol{Y}, \boldsymbol{W}) = \boldsymbol{T} = \boldsymbol{Y} + \boldsymbol{G}\boldsymbol{W}$
Initialization: $\boldsymbol{W} = 0$, $\boldsymbol{T} = \boldsymbol{Y} + \boldsymbol{G}\boldsymbol{W}$, $\beta > 0$, $\lambda > 0$, $0 \leq w \leq 1$;

$$\sigma^2 = \frac{1}{DNM} \sum_{n=1}^{N} \sum_{m=1}^{M} \|\boldsymbol{x}_n - \boldsymbol{y}_m\|^2;$$

Construct: $\boldsymbol{G}_{ij} = \exp(-\frac{1}{2\beta^2}\|\boldsymbol{y}_i - \boldsymbol{y}_j\|^2)$;

while *not converged* **do**

E-step: Compute $\boldsymbol{P}$;

$$p_{mn} = \frac{\exp(-\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \boldsymbol{T}_m\|^2)}{\sum_{k=1}^{M} \exp(-\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \boldsymbol{T}_k\|^2) + (2\pi\sigma^2)^{D/2} \frac{w}{1-w} \frac{M}{N}};$$

M-step: Solve for $\boldsymbol{W}, \sigma^2$;

$$(\boldsymbol{G} + \lambda\sigma^2 d(\boldsymbol{P}\mathbf{1})^{-1})\boldsymbol{W} = d(\boldsymbol{P}\mathbf{1})^{-1}\boldsymbol{P}\boldsymbol{X} - \boldsymbol{Y};$$

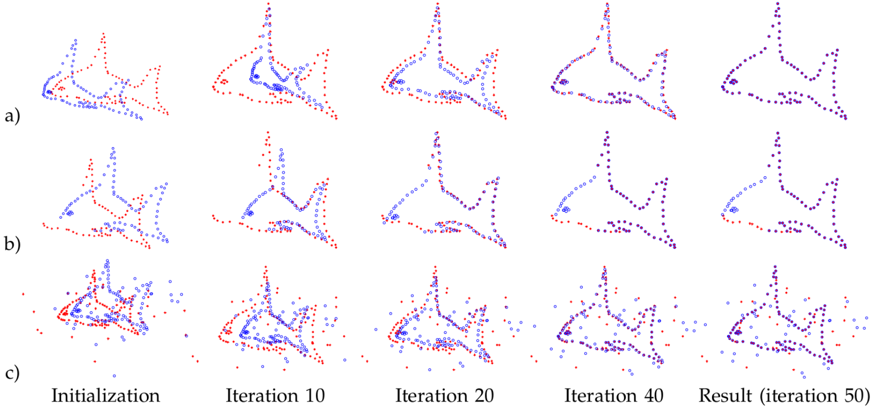
$$N_P = \mathbf{1}^T \boldsymbol{P}\mathbf{1};$$

$$\boldsymbol{T} = \boldsymbol{Y} + \boldsymbol{G}\boldsymbol{W};$$

$$\sigma^2 = \frac{1}{N_P D}(\mathrm{tr}[\boldsymbol{X}^T d(\boldsymbol{P}^T\mathbf{1})\boldsymbol{X}] - 2 \cdot \mathrm{tr}[(\boldsymbol{P}\boldsymbol{X})^T \boldsymbol{T}] + \mathrm{tr}[\boldsymbol{T}^T d(\boldsymbol{P}\mathbf{1})\boldsymbol{T}]);$$

end

- The range of filtered frequencies (smoothness) can be controlled by β
- The Tikonov regulatizer (norm of the deformation) can be controlled by λ


a)
b)
c)
Initialization
Iteration 10
Iteration 20
Iteration 40
Result (iteration 50)

Advantages:

1. Simple robust formulation
2. Works on Point Clouds and Meshes
3. Explicitly models outliers using probability w
4. Can be implemented on the GPU

Disadvantages:

1. Parameters λ and β control the smoothness of the deformation field
2. Acceleration only works for Gaussian kernel functions
3. Unknown if always converges (until 2020, Bayesian Coherent point drift)

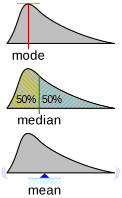
Variational Bayesian Inference (VBI)

Idea: Formulate CPD in the Bayesian setting, replacing the motion coherence theory of CPD and replace it with Bayesian inference.

- Introduce motion coherence using a prior distribution of displacement vectors
- Convergence is guaranteed
- Parameters of motion coherence can be interpreted intuitively
- Combines rigid CPD and non-rigid CPD
- Can be accelerated with non-Gaussian kernel functions

Reminder: Bayesian inference estimates a set of unobserved variables $\boldsymbol{\theta}$ from a set of observations $\boldsymbol{X}$ by computing the mode of a posterior $p(\boldsymbol{\theta}|\boldsymbol{X})$ or the expectation of $\boldsymbol{\theta}$ over $p(\boldsymbol{\theta}|\boldsymbol{X})$.

- The mode of a probability distribution is the maximum likelihood
- The mean can be computed using the expectation $\mathbb{E}_p[x]$ w.r.t. $p(x)$
- For a gaussian, the mode, the median and the mean fall onto the same x



In the Bayesian setting we need to provide prior distributions for all parameters.

Reminder: Bayes' theorem

$$p(\boldsymbol{\theta}|\boldsymbol{X}) = \frac{p(\boldsymbol{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\boldsymbol{X})}$$

For a general probability distribution $p(\boldsymbol{X}|\boldsymbol{\theta})$

- the product $p(\boldsymbol{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})$ can have a different functional form than the prior $p(\boldsymbol{\theta})$

We can seek a prior $p(\boldsymbol{\theta})$ that is conjugate to the likelihood function, so that the posterior distribution has the same functional form as the prior.

Choosing a conjugate prior is an algebraic convenience, giving a closed-form expression for the posterior. Otherwise numerical integration may be necessary.

All members of the exponential family have conjugate priors.

Variational techniques (VBI) are typically used to form an approximation for

$$p(\boldsymbol{\theta}|\boldsymbol{X}) = \frac{p(\boldsymbol{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\boldsymbol{X})} = \frac{p(\boldsymbol{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int_{\boldsymbol{\theta}} p(\boldsymbol{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})}$$

The marginalization over $\boldsymbol{\theta}$ to calculate $p(\boldsymbol{X})$ in the denominator is typically intractable, because, e.g. the search space of $\boldsymbol{\theta}$ is combinatorially large.

Therefore, we seek an approximation using a so-called **variational distribution** $q(\boldsymbol{\theta}) \approx p(\boldsymbol{\theta}|\boldsymbol{X})$.

The distribution of $q(\boldsymbol{\theta})$ is restricted to belong to a family of distributions of simpler form, e.g. a family of Gaussian distributions than $p(\boldsymbol{\theta}|\boldsymbol{X})$.

The dissimilarity between distributions is measured using a function $D(\cdot||\cdot)$ and hence inference is performed by selecting the distribution $q(\boldsymbol{\theta})$ which minimize the dissimilarity. The most common type of VBI uses the Kullback-Leibler divergence to measure the dissimilarity between distributions q and p

$$D_{KL}(q||p) = \int q(\boldsymbol{\theta}) \ln\left(\frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta}|\boldsymbol{X})}\right) d\boldsymbol{\theta}$$

Given that

$$p(\boldsymbol{\theta}|\boldsymbol{X}) = \frac{p(\boldsymbol{X}, \boldsymbol{\theta})}{p(\boldsymbol{X})}$$

the Kullback-Leibler divergence can be written as

$$\begin{aligned} D_{KL}(q||p) &= \int q(\boldsymbol{\theta}) \left[\ln \left(\frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta}, \boldsymbol{X})} \right) + \ln p(\boldsymbol{X}) \right] d\boldsymbol{\theta} \\ &= \int q(\boldsymbol{\theta}) [\ln q(\boldsymbol{\theta}) - \ln p(\boldsymbol{\theta}, \boldsymbol{X})] d\boldsymbol{\theta} + \int q(\boldsymbol{\theta}) \ln p(\boldsymbol{X}) d\boldsymbol{\theta} \end{aligned}$$

Because $q(\boldsymbol{X})$ is constant w.r.t $\boldsymbol{\theta}$ and $\int q(\boldsymbol{\theta}) d\boldsymbol{\theta} = 1$ we have

$$D_{KL}(q||p) = \int q(\boldsymbol{\theta}) [\ln q(\boldsymbol{\theta}) - \ln p(\boldsymbol{\theta}, \boldsymbol{X})] d\boldsymbol{\theta} + \ln p(\boldsymbol{X})$$

which can be written using the definition of the expected value as

$$D_{KL}(q||p) = \mathbb{E}_q [\ln q(\boldsymbol{\theta}) - \ln p(\boldsymbol{\theta}, \boldsymbol{X})] + \ln p(\boldsymbol{X})$$

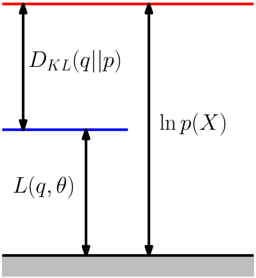
We can rearrange for the log evidence as

$$\ln p(\boldsymbol{X}) = D_{KL}(q||p) - \mathbb{E}_q[\ln q(\boldsymbol{\theta}) - \ln p(\boldsymbol{\theta}, \boldsymbol{X})] = D_{KL}(q||p) + L(q)$$

The **evidence lower bound** $L(q) = \mathbb{E}_q[\ln p(\boldsymbol{\theta}, \boldsymbol{X})] - \mathbb{E}_q[\ln p(\boldsymbol{\theta})]$ is known as the **variational free energy**, because is can be expressed as a negative "energy" $\mathbb{E}_q[\ln p(\boldsymbol{\theta}, \boldsymbol{X})]$ plus the entropy of q, in analogy with thermodynamic free energy.

VBI is then defined as the the maximization of the Lower bound which is equivalent to the minimization of the Kullback-Leibler divergence.

$$\begin{aligned} L(q) &= -\int q(\boldsymbol{\theta})[\ln q(\boldsymbol{\theta}) - \ln p(\boldsymbol{\theta}, \boldsymbol{X})]d\boldsymbol{\theta} \\ &= \int q(\boldsymbol{\theta})[\ln p(\boldsymbol{\theta}, \boldsymbol{X}) - \ln q(\boldsymbol{\theta})]d\boldsymbol{\theta} \\ &= \int q(\boldsymbol{\theta}) \ln\left(\frac{p(\boldsymbol{\theta}, \boldsymbol{X})}{q(\boldsymbol{\theta})}\right)d\boldsymbol{\theta} \end{aligned}$$



To avoid the trivial solution $q(\boldsymbol{\theta}) = p(\boldsymbol{\theta}|\boldsymbol{X})$ we need to impose some constraints on this optimization problem.

Suppose that $\boldsymbol{\theta}$ can be divided into J groups, i.e. $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_J)$ and denote the marginal distribution of $\boldsymbol{\theta}_j$ w.r.t $q(\boldsymbol{\theta})$ by $q_j(\boldsymbol{\theta}_j)$. Constrain the posterior distribution $q(\boldsymbol{\theta})$ to be the product of its marginals

$$q(\boldsymbol{\theta}) = \prod_{j=1}^{J} q_j(\boldsymbol{\theta}_j)$$

This factorized form of variational inference corresponds to an approximation framework called **mean field theory**. Assume that only q_i is unknown among the factorized distributions

$$\{q_1, \dots, q_J\}$$

The lower bound $L(q)$ with the factorized $q(\boldsymbol{\theta})$ is then

$$
\begin{aligned}
L(q) &= \int \prod_i q_i(\boldsymbol{\theta}_i) \left(\ln p(\boldsymbol{X}, \boldsymbol{\theta}) - \sum_i \ln q_i(\boldsymbol{\theta}_i) \right) d\boldsymbol{\theta} \\
&= \int q_j(\boldsymbol{\theta}_j) \underbrace{\left(\int \left(\prod_{i \neq j} q_i(\boldsymbol{\theta}_i) \right) \ln p(\boldsymbol{X}, \boldsymbol{\theta}) d\boldsymbol{\theta}_i \right)}_{\mathbb{E}_{i \neq j}[\ln p(\boldsymbol{X}, \boldsymbol{\theta})] =: \ln \tilde{q}(\boldsymbol{X}, \boldsymbol{\theta})} d\boldsymbol{\theta}_j - \int q_j(\boldsymbol{\theta}_j) \ln q_j(\boldsymbol{\theta}_j) d\boldsymbol{\theta}_j + \text{const} \\
&= \int q_j(\boldsymbol{\theta}_j) \ln \tilde{q}(\boldsymbol{X}, \boldsymbol{\theta}_j) d\boldsymbol{\theta}_j - \int q_j(\boldsymbol{\theta}_j) \ln q_j(\boldsymbol{\theta}_j) d\boldsymbol{\theta}_j + \text{const}
\end{aligned}
$$

with $\ln \tilde{q}(\boldsymbol{X}, \boldsymbol{\theta}_j) = \mathbb{E}_{i \neq j}[\ln p(\boldsymbol{X}, \boldsymbol{\theta})] + \text{const}$

Keep the $\{q_{i \neq j}\}$ fixed and maximize $L(q)$ w.r.t all possible forms of the distribution $q_j(\boldsymbol{\theta}_j)$.
Recognize that the lower bound $L(q)$ is now a negative Kullback-Leibler divergence between $q_j(\boldsymbol{\theta}_j)$ and $\tilde{q}(\boldsymbol{X}, \boldsymbol{\theta}_j)$ and maximized as

$$q_j^\star(\boldsymbol{\theta}_j) = \arg\max_{q_j} \int q_j(\boldsymbol{\theta}_j) \ln\left(\frac{\exp(\mathbb{E}_{i \neq j}[\ln p(\boldsymbol{X}, \boldsymbol{\theta})])}{q_j(\boldsymbol{\theta}_j)}\right) d\boldsymbol{\theta}_j$$

where $\mathbb{E}_{i \neq j}[\ln p(\boldsymbol{X}, \boldsymbol{\theta})] = \int \ln p(\boldsymbol{X}, \boldsymbol{\theta}) \prod_{j \neq i} q_j(\boldsymbol{\theta}_j) d\boldsymbol{\theta}_j$.

The general solution of this negative Kullback-Leibler divergence is then

$$\ln q_j^\star(\boldsymbol{\theta}_j) = \mathbb{E}_{i \neq j}[\ln p(\boldsymbol{X}, \boldsymbol{\theta})] + \text{const}\,.$$

where the constant term is the normalization constant of $q_j^\star(\boldsymbol{\theta}_j)$.

Important: This form is very helpful, because we can just plug-in the joint distribution and derive update rules for the individual parameters from there.

Taking the exponential on both sides and normalize we have

$$q_j^\star(\boldsymbol{\theta}_j) = \frac{\exp(\mathbb{E}_{i\neq j}[\ln p(\boldsymbol{X}, \boldsymbol{\theta})])}{\int \exp(\mathbb{E}_{i\neq j}[\ln p(\boldsymbol{X}, \boldsymbol{\theta})]) d\boldsymbol{\theta}_j}$$

Suppose that $\tilde{q}(\boldsymbol{\theta}) = q_j^\star(\boldsymbol{\theta}_j) \prod_{j\neq i} q_i(\boldsymbol{\theta}_i)$, then the inequality of the lower bounds $L(\tilde{q}) \geq L(q)$ always holds and is tight if and only if $q_j^\star(\boldsymbol{\theta}_j) = q_j(\boldsymbol{\theta}_j)$.

A solution for the optimization problem is therefore obtained by

1. Initialize q_i for all $i \in \{1, \ldots, J\}$
2. Cycle the update of q_j with the other q_i fixed for each $i \in \{1, \ldots, J\}$
3. Repeat step 2 until convergence

Convergence is guaranteed because $L(q)$ is monotonically increasing and bounded because of the inequalities $\ln p(\boldsymbol{X}) \geq L(q^\star) \geq L(\tilde{q}) \geq L(q)$[5].

[5] C. M. Bishop. *Pattern recognition and machine learning*. springer, 2006.

We illustrate the factorized variational approximation using a Gaussian distribution over a single variable x. Our goal is to infer the posterior distribution for the mean μ and the precision τ, given a data set $\mathcal{D} = \{x_1, \ldots, x_N\}$ of observed values of x which are assumed to be drawn independently from the Gaussian.
The likelihood function is given by

$$p(\mathcal{D}|\mu, \tau) = \left(\frac{\tau}{2\pi}\right)^{N/2} \exp\left(-\frac{\tau}{2}\sum_{n=1}^{N}(x_n - \mu)^2\right)$$

We now introduce conjugate prior distributions for μ and τ given by

$$p(\mu|\tau) = \mathcal{N}(\mu|\mu_0, (\lambda_0\tau)^{-1})$$
$$p(\tau) = \mathrm{Gam}(\tau|a_0, b_0) = \frac{1}{\Gamma(a_0)} b_0^{a_0} \tau^{a_0 - 1} \exp(-b_0\tau)$$

Consider a factorized variational approximation to the posterior distribution given by

$$q(\mu, \tau) = q_\mu(\mu) q_\tau(\tau)$$

The optimum factors $q_\mu(\mu)$ and $q_\tau(\tau)$ can be obtained from the general solution of VBI as follows

$$\begin{aligned}\ln q_\mu^\star(\mu) &= \mathbb{E}_\tau[\ln p(\mathcal{D}|\mu,\tau) + \ln p(\mu|\tau)] + \text{const} \\ &= -\frac{\mathbb{E}_\tau[\tau]}{2}\left(\lambda_0(\mu-\mu_0)^2 + \sum_{n=1}^{N}(x_n-\mu)^2\right) + \text{const}\end{aligned}$$

Completing the square over μ we see that $q_\mu(\mu)$ is a Gaussian $\mathcal{N}(\mu|\mu_N, \lambda_N^{-1})$ with mean and precision given by

$$\begin{aligned}\mu_N &= \frac{\lambda_0\mu_0 + N\bar{x}}{\lambda_0 + N} \\ \lambda_N &= (\lambda_0 + N)\mathbb{E}_\tau[\tau]\end{aligned}$$

Note that for $N \to \infty$ this gives the maximum likelihood result in which $\mu_N = \bar{x}$ and the precision is infinite.

Similarly, the optimal solution for the factor $q_\tau(\tau)$ is given by

$$\begin{aligned}\ln q_\tau^\star(\tau) &= \mathbb{E}_\mu[\ln p(\mathcal{D}|\mu,\tau) + \ln p(\mu|\tau))] + \ln p(\tau) + \text{const} \\ &= (a_0 - 1)\ln\tau - b_0\tau + \frac{N}{2}\ln\tau - \frac{\tau}{2}\mathbb{E}_\mu[\lambda_0(\mu-\mu_0)^2 + \sum_{n=1}^{N}(x_n-\mu)^2] + \text{const}\end{aligned}$$

and hence $q_\tau(\tau)$ is a gamma distribution $\mathrm{Gam}(\tau|a_N, b_N)$ with

$$\begin{aligned}a_N &= a_0 + \frac{N}{2} \\ b_N &= b_0 + \frac{1}{2}\mathcal{E}_\mu\left[\lambda_0(\mu-\mu_0)^2 + \sum_{n=1}^{N}(x_n-\mu)^2\right]\end{aligned}$$

Again we get the expected behavior when $N \to \infty$.

- These specific functional forms for q_μ and q_τ were not assumed and arose naturally from the structure of the likelihood and the corresponding priors.
- We have expressions for the optimal distributions q_μ and q_τ each of which depends on moments evaluated w.r.t. the other distribution.
- One approach to finding a solution is therefore to make an initial guess, for say the moment $\mathbb{E}_\tau[\tau]$ and use this to re-compute the distribution q_μ.
- Given this revised distribution we can extract the required moments $\mathbb{E}_\mu[\mu]$ and $\mathbb{E}_\mu[\mu^2]$, and use these to re-compute the distribution q_τ.

Note: Since the space of hidden variables here is only two dimensional, we can illustrate the variational approximation to the posterior distribution by protting contours off both the true posterior and the factorized approximation (next slilde).

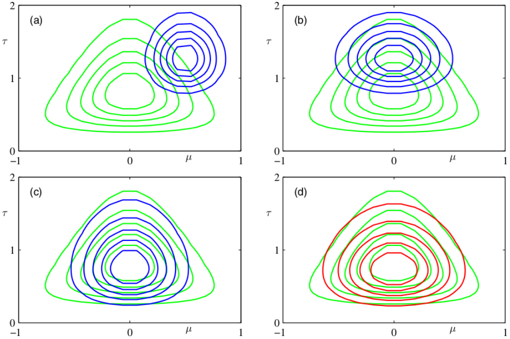


Illustration of variational inference for the mean μ and precision τ of a univariate Gaussian distribution. Contours of the true posterior distribution are shown in green.
(a) Contours of the initial factorized approximation $q_\mu(\mu)q_\tau(\tau)$ are shown in blue.
(b) After re-estimating the factor $q_\mu(\mu)$.
(c) After re-estimating the factor $q_\tau(\tau)$.
(d) Contours of the optimal factorized approximation, to which the iterative scheme converges, are shown in red.

Bayesian Coherent-Point-Drift

CPD-Assumptions: Deformation model

Define the non-rigid transformation model $\mathcal{T}$ as a combination of a similarity transform T and a non-rigid deformation[6]:

$$\mathcal{T}(\boldsymbol{y}_m) = T(\boldsymbol{y}_m + \boldsymbol{v}_m) = s \cdot \boldsymbol{R}(\boldsymbol{y}_m + \boldsymbol{v}_m) + \boldsymbol{t}$$

where

- $s \in \mathbb{R}$ is a scale factor
- $\boldsymbol{R} \in \mathbb{R}^{D \times D}$ is a rotation matrix
- $\boldsymbol{t} \in \mathbb{R}^{D}$ is a translation vector
- $\boldsymbol{v}_m \in \mathbb{R}^{D}$ is a displacement vector

[6] O. Hirose. "A Bayesian Formulation of Coherent Point Drift". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2020), pp. 1–1.

Under the assumptions

1. A target point $\boldsymbol{x}_n$ is selected as either a point forming a target shape or an outlier with outlier probability w
2. If $\boldsymbol{x}_n$ is selected as an outlier, $\boldsymbol{x}_n$ is generated from an outlier distribution $p_{out}(\boldsymbol{x}_n)$
3. If $\boldsymbol{x}_n$ is a non-outlier, an index $m \in \{1, \ldots, M\}$ indicating that $\boldsymbol{x}_n$ corresponds to $\boldsymbol{y}_m$, is sampled with a probability α_m, where $\sum_m^M \alpha_m = 1$
4. Then $\boldsymbol{x}_n$ is generated from a D-dimensional multivariate normal distribution with a mean vector $\mathcal{T}(\boldsymbol{y}_m)$ and a covariance matrix $\sigma^2 \boldsymbol{I}_D$

The GMM that generates the target point cloud is defined as

$$\phi_{mn} = \phi(\boldsymbol{x}_n | \mathcal{T}(\boldsymbol{y}_m), \sigma^2 \boldsymbol{I}_D) = \frac{1}{\sqrt{\det(2\pi\sigma^2 \boldsymbol{I}_D)}} \exp(-\frac{1}{2\sigma^2} \|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{y}_m)\|^2)$$

for $\boldsymbol{x}_n$ when event $e_n = m$ that point n is a non-outlier and in correspondence with point m.

The GMM is defined with an explicit definition of unknown correspondences, i.e.

- c_n is an indicator value if the nth target point is a non-outlier of a source shape
- e_n is an index value representing that the nth target point corresponds to the mth source point if $e_n = m$

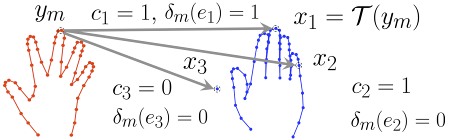


Figure: Notation with respect to corresponding points. The point sets colored red and blue represent the source point set Y and the target point set X, respectively. The variable $c_n \in \{0, 1\}$ indicates whether or not x_n is a non-outlier, whereas the variable $e_n \in \{1, \cdots, M\}$ specifies the index of a point in Y that corresponds to x_n. Target points x_1, x_2, and x_3 represent the point that corresponds to y_m, a non-outlier that does not correspond to y_m, and an outlier, respectively. (Taken from Hirose, 2020.)

The joint distribution of $(\boldsymbol{x}_n, c_n, e_n)$ given $(\boldsymbol{y}, \boldsymbol{v}, \boldsymbol{\alpha}, \rho, \sigma^2)$ is defined as a two-component mixture distribution and an M-component mixture distribution

$$p(\boldsymbol{x}_n, e_n, c_n | \boldsymbol{y}, \boldsymbol{v}, \boldsymbol{\alpha}, \rho, \sigma^2) = (w p_{out}(\boldsymbol{x}_n))^{1-c_n} \left((1-w) \prod_{m=1}^{M} (\alpha_m \phi_{mn})^{\delta_m(e_n)} \right)^{c_n}$$

Define the prior $p(\boldsymbol{\alpha})$ as a Dirichlet distribution

$$p(\boldsymbol{\alpha}) = \mathrm{Dir}(\boldsymbol{\alpha} | \kappa \mathbf{1}_M) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{m=1}^{M} \alpha_m^{\kappa - 1}$$

where $\kappa > 0$ is the parameter that controls the shape of the Dirichlet distribution , $\mathbf{1}_M$ is the vector of all 1s of size M and $B(\boldsymbol{\alpha})$ is the normalizing constant.

Motion Coherence is defined as a prior distribution of displacement vectors. Suppose G is a positive kernel matrix with $g_{ij} = \mathcal{K}(\boldsymbol{y}_i, \boldsymbol{y}_j)$, $\mathcal{K}(\cdot, \cdot)$ positive-definite kernel, then the prior of $\boldsymbol{v}$ is the following Normal distribution

$$p(\boldsymbol{v}|\boldsymbol{y}) = \phi(\boldsymbol{v}|\boldsymbol{0}, \lambda^{-1}\boldsymbol{G} \otimes \boldsymbol{I}_D)$$

where λ is a positive constant and $\otimes$ is the Kronecker product. This prior encodes motion coherence because displacement vectors $\boldsymbol{v}_i$ and $\boldsymbol{v}_j$ correlate with each other if the affinity value $\lambda^{-1}g_{ij}$ is sufficiently large.

For the simplicity of the generative model there is no prior over the

- source point set $\boldsymbol{y}$
- residual variance σ^2
- variables ρ of the similarity transform T

Full Joint Distribution

The full joint distribution for the whole target point set with priors is then

$$p(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\theta}) \propto p(\boldsymbol{v}|\boldsymbol{y})p(\boldsymbol{\alpha}) \prod_{n=1}^{N} p(\boldsymbol{x}_n, c_n, e_n|\boldsymbol{y}, \boldsymbol{v}, \boldsymbol{\alpha}, \sigma^2)$$

where $\boldsymbol{\theta} = (\boldsymbol{v}, \boldsymbol{\alpha}, \boldsymbol{c}, \boldsymbol{e}, \rho, \sigma^2)$ is the set of latent variables that mediate the generation of $\boldsymbol{x}$ given $\boldsymbol{y}$.

- replaces point set registration with a probabilistic density estimation
- the exact computation of the estimate using expectation of $\boldsymbol{\theta}$ over $p(\boldsymbol{\theta}|\boldsymbol{x}, \boldsymbol{y})$ or the maximum mode of $p(\boldsymbol{\theta}|\boldsymbol{x}, \boldsymbol{y})$ is roughly $(M+1)^N$ with a naive method.
- therefore derive an algorithm for estimating a reasonable $\boldsymbol{\theta}$ using **Variational Bayesian Inference**

Approximate $p(\boldsymbol{\theta}|\boldsymbol{x},\boldsymbol{y})$ using a factorized

$$q(\boldsymbol{\theta}) = q_1(\boldsymbol{v},\boldsymbol{\alpha})q_2(\boldsymbol{c},\boldsymbol{e})q_3(\rho,\sigma^2)$$

Assume that $q_3(\rho,\sigma^2) = \delta(\rho,\sigma^2)$ for simplification, where δ without subscript represents a Dirac delta function, i.e. q_3 is the distribution with a point mass at (ρ,σ^2).

This assumption means that q_3 is characterized by the fist moment only and the second and higher moments can be represented as the product of the first moment.

The maximum mode or expectation regarding q will be available after performing iterative updates of q_1, q_2 and q_3 until guaranteed convergence.

To compute the optimal $q_1^\star(\boldsymbol{v}, \boldsymbol{\alpha})$ we fix $q_2(\boldsymbol{c}, \boldsymbol{e})q_3(\rho, \sigma^2)$ and put the full joint-distribution $p(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\theta})$ into the general solution of VBI

$$\begin{aligned}
\ln q_1^\star(\boldsymbol{v}, \boldsymbol{\alpha}) &= \mathbb{E}_{q_2 q_3}[\ln p(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\theta})] + \text{const.} \\
&= \mathbb{E}_{q_2 q_3}\left[\ln\left(p(\boldsymbol{v}|\boldsymbol{y})p(\boldsymbol{\alpha})\prod_{n=1}^{N} p(\boldsymbol{x}_n, c_n, e_n|\boldsymbol{y}, \boldsymbol{v}, \boldsymbol{\alpha}, \rho, \sigma^2)\right)\right] + \text{const.} \\
&= \mathbb{E}_{q_2 q_3}\left[\ln p(\boldsymbol{v}|\boldsymbol{y}) + \ln p(\boldsymbol{\alpha}) + \sum_{n=1}^{N} \ln p(\boldsymbol{x}_n, c_n, e_n|\boldsymbol{y}, \boldsymbol{v}, \boldsymbol{\alpha}, \rho, \sigma^2)\right] + \text{const.}
\end{aligned}$$

Since $p(\boldsymbol{v}|\boldsymbol{y})$ and $p(\boldsymbol{\alpha})$ are independent w.r.t. the parameters of $q_2 q_3$ and we have

$$\begin{aligned}
\ln q_1^\star(\boldsymbol{v}, \boldsymbol{\alpha}) &= \ln p(\boldsymbol{v}|\boldsymbol{y}) + \ln p(\boldsymbol{\alpha}) + \sum_{n=1}^{N} \mathbb{E}_{q_2 q_3}\left[\ln p(\boldsymbol{x}_n, c_n, e_n|\boldsymbol{y}, \boldsymbol{v}, \boldsymbol{\alpha}, \rho, \sigma^2)\right] \\
&= -\frac{\lambda}{2}\boldsymbol{v}^T\tilde{\boldsymbol{G}}^{-1}\boldsymbol{v} + \sum_{m=1}^{M} \ln \alpha_m^{\kappa-1} + \sum_{n=1}^{N}\sum_{m=1}^{M} \mathbb{E}_{q_2 q_3}[\delta_m(e_n)c_n \ln(\alpha_m \phi_{mn})] + \text{const.}
\end{aligned}$$

The last term can be derived by using the linearity of the Expected value and the application of $\ln(a^b) = a\ln(b)$.

The right hand side of the equation can be decomposed into a sum of terms including only $\boldsymbol{\alpha}$ and those including only v, which implies that $q_1^\star(\boldsymbol{v}, \boldsymbol{\alpha})$ is factorized into its marginals, i.e. $q_1^\star(\boldsymbol{v}, \boldsymbol{\alpha}) = q_{\boldsymbol{\alpha}}^\star(\boldsymbol{\alpha}) q_{\boldsymbol{v}}^\star(\boldsymbol{v})$.

Therefore the update of $q_1^\star(\boldsymbol{v}, \boldsymbol{\alpha})$ is divided into two steps:

1. update of $q_{\boldsymbol{\alpha}}^\star(\boldsymbol{\alpha})$
2. update of $q_{\boldsymbol{v}}^\star(\boldsymbol{v})$

First we show that $q_{\boldsymbol{\alpha}}^\star(\boldsymbol{\alpha})$ is a **Dirichlet distribution**, and then we show that $q_{\boldsymbol{v}}^\star(\boldsymbol{v})$ follows a **multivariate normal distribution**.

We extract the α_m-term from the right hand sum using the linearity of the Expected value and define $p_{mn} = \mathbb{E}_{q2q3}[c_n \delta_m(e_n)]$

$$\begin{aligned}
\sum_{n=1}^{N} \sum_{m=1}^{M} \mathbb{E}_{q_2 q_3}[c_n \delta_m(e_n) \ln(\alpha_m \phi_{mn})] &= \sum_{n=1}^{N} \sum_{m=1}^{M} \mathbb{E}_{q_2 q_3}[c_n \delta_m(e_n)](\ln(\alpha_m) + \ln(\phi_{mn})) \\
&= \sum_{n=1}^{N} \sum_{m=1}^{M} p_{mn}(\ln(\alpha_m) + \ln(\phi_{mn})) \\
&= \sum_{m=1}^{M} \left(\sum_{n=1}^{N} p_{mn} \right) \ln \alpha_m + \sum_{m,n}^{M,N} p_{mn} \ln \phi_{mn}
\end{aligned}$$

Suppose that $\nu_m = \sum_{n=1}^{N} p_{mn}$, which is interpreted as the estimated number of points in a target point set that matches $\boldsymbol{y}_m$ we get for the terms containing α_m

$$\ln q^\star_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}) = \sum_{m=1}^{M} \ln \alpha_m^{\kappa-1} + \sum_{m=1}^{M} \nu_m \ln \alpha_m + \text{const.} = \sum_{m=1}^{M} \ln \alpha^{\kappa-1+\nu_m} + \text{const.}$$

by applying $\ln(x) + \ln(y) = \ln(xy)$ repeatedly, leading to

$$q^\star_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}) = \mathrm{Dir}(\alpha|\kappa \mathbf{1}_M + \boldsymbol{\nu})$$

The closed form expression for $q^\star_{\boldsymbol{v}}(\boldsymbol{v})$ is formed by focusing on the terms involving $\boldsymbol{v}$. We insert $\phi_{mn} = \frac{1}{\sqrt{\det\left(2\pi\sigma^2 \boldsymbol{I}_D\right)}} \exp\left(-\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{y}_m)\|^2\right)$ into our equation and get

$$\ln q^\star_{\boldsymbol{v}}(\boldsymbol{v}) = -\frac{\lambda}{2}\boldsymbol{v}^T \tilde{G}^{-1}\boldsymbol{v} - \frac{1}{2\sigma^2}\sum_{m=1}^{M}\sum_{n=1}^{N} p_{mn}\|\boldsymbol{x}_n - T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2 + \text{const.}$$

Let us define the symbols $\tilde{\boldsymbol{P}} = \boldsymbol{P} \otimes \boldsymbol{I}_D$, $\tilde{\boldsymbol{\nu}} = \boldsymbol{\nu} \otimes \mathbf{1}_D$, $\tilde{\boldsymbol{R}} = \boldsymbol{I}_M \otimes \boldsymbol{R}$ and $\tilde{T}(\boldsymbol{y} + \boldsymbol{v}) = s\tilde{\boldsymbol{R}}(\boldsymbol{y} + \boldsymbol{v}) + (\mathbf{1}_M \otimes \boldsymbol{t})$. Also denote the inverse function of $\tilde{T}$ by $\tilde{T}^{-1}$ and using the equations $\tilde{\boldsymbol{R}}d(\tilde{\boldsymbol{\nu}}) = d(\tilde{\boldsymbol{\nu}})\tilde{\boldsymbol{R}}$ and $\boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}_D$, the second terms ignoring terms without v and $-\frac{1}{2\sigma^2}$ is

$$\begin{aligned}
\sum_{n,m}^{N,M} p_{mn}\|\boldsymbol{x}_n - T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2 &= \sum_{n,m}^{N,M} p_{mn}\left(\|T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2 - 2\boldsymbol{x}_n^T T(\boldsymbol{y}_m + \boldsymbol{v}_m)\right) + \text{const.} \\
&= \tilde{T}(\boldsymbol{y} + \boldsymbol{v})^T d(\tilde{\boldsymbol{\nu}})\tilde{T}(\boldsymbol{y} + \boldsymbol{v}) - 2\boldsymbol{x}^T\tilde{\boldsymbol{P}}^T\tilde{T}(\boldsymbol{y} + \boldsymbol{v}) + \text{const.} \\
&= s^2\boldsymbol{v}^T d(\tilde{\nu})\boldsymbol{v} - 2s(\hat{\boldsymbol{x}} - \tilde{T}(\boldsymbol{y}))^T d(\tilde{\boldsymbol{\nu}})\tilde{\boldsymbol{R}}(\boldsymbol{y} + \boldsymbol{v}) + \text{const.} \\
&= s^2\boldsymbol{v}^T d(\tilde{\boldsymbol{\nu}})\boldsymbol{v} - 2s^2(\tilde{T}^{-1}(\hat{\boldsymbol{x}}) - \boldsymbol{y})^T d(\tilde{\boldsymbol{\nu}})\boldsymbol{v} + \text{const.}
\end{aligned}$$

We use the symbol $\hat{\boldsymbol{x}} = d(\tilde{\boldsymbol{\nu}})^{-1}\tilde{\boldsymbol{P}}\boldsymbol{x}$ representing the shape for which $\tilde{T}^{-1}(\hat{\boldsymbol{x}})$ is interpreted as the inverse alignment from $\boldsymbol{x}$ to $\boldsymbol{y}$. We recognize that the term in the exponential of a normal distribution can be expanded as a quadratic form

$$-\frac{1}{2}(\boldsymbol{v}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{v}-\boldsymbol{\mu}) = -\frac{1}{2}\left(\boldsymbol{v}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{v} - 2\boldsymbol{v}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}\right) + \text{const.}$$

Then we can reformulate $\ln q_{\boldsymbol{v}}^{\star}(\boldsymbol{v})$ as

$$\begin{aligned}
\ln q_{\boldsymbol{v}}^{\star}(\boldsymbol{v}) &= -\frac{\lambda}{2}\boldsymbol{v}^T\tilde{\boldsymbol{G}}^{-1}\boldsymbol{v} - \frac{1}{2\sigma^2}\left(s^2\boldsymbol{v}^T d(\tilde{\boldsymbol{\nu}})\boldsymbol{v} - 2s^2(\tilde{T}^{-1}(\hat{\boldsymbol{x}})-\boldsymbol{y})^T d(\tilde{\boldsymbol{\nu}})\boldsymbol{v}\right) + \text{const.} \\
&= -\frac{1}{2}\left(\boldsymbol{v}^T\underbrace{(\lambda\tilde{\boldsymbol{G}}^{-1} + \frac{s^2}{\sigma^2}d(\tilde{\boldsymbol{\nu}}))}_{\tilde{\boldsymbol{\Sigma}}^{-1}}\boldsymbol{v} - 2(\boldsymbol{v}^T\tilde{\boldsymbol{\Sigma}}^{-1}\underbrace{\frac{s^2}{\sigma^2}\tilde{\boldsymbol{\Sigma}}d(\tilde{\boldsymbol{\nu}})(\tilde{T}^{-1}(\hat{\boldsymbol{x}}))-\boldsymbol{y})}_{\tilde{\boldsymbol{\mu}}}\right) + \text{const.} \\
&= -\frac{1}{2}(\boldsymbol{v}-\tilde{\boldsymbol{\mu}})^T\tilde{\boldsymbol{\Sigma}}^{-1}(\boldsymbol{v}-\tilde{\boldsymbol{\mu}}) + \text{const.}
\end{aligned}$$

where $\tilde{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}\otimes\boldsymbol{I}_D$ with $\boldsymbol{\Sigma} = (\lambda\boldsymbol{G}^{-1} + \frac{s^2}{\sigma^2}d(\boldsymbol{\nu}))^{-1} \in \mathbb{R}^{M\times M}$.

Taking the exponential on both sides and dividing by the normalization constant we obtain the posterior distribution $q^\star_{\boldsymbol{v}}(\boldsymbol{v})$ as a multivariate normal distribution

$$q^\star_{\boldsymbol{v}}(\boldsymbol{v}) = \phi(v|\frac{s^2}{\sigma^2}\tilde{\boldsymbol{\Sigma}}d(\tilde{\boldsymbol{\nu}})(\tilde{T}^{-1}(\hat{\boldsymbol{x}}) - \boldsymbol{y}), \tilde{\boldsymbol{\Sigma}})$$

This gives some insights into motion coherence. Because $T^{-1}(\hat{\boldsymbol{x}_m}) - \boldsymbol{y}_m$ is interpreted as the residual for the mth source point, the meaning of the mean vector $\hat{\boldsymbol{v}} = \frac{s^2}{\sigma^2}\tilde{\boldsymbol{\Sigma}}d(\tilde{\boldsymbol{\nu}})(\tilde{T}^{-1}(\hat{\boldsymbol{x}}) - \boldsymbol{y})$ can be interpreted as follows:

- If λ is sufficiently large, $\hat{\boldsymbol{v}}_m \approx \lambda^{-1}\sum_{m'=1}^{M} \boldsymbol{g}_{mm'}\boldsymbol{\nu}_{m'}(T^{-1}(\hat{\boldsymbol{x}}_{m'}) - \boldsymbol{y}_{m'})$, i.e. it is computed by smoothing the weighted residual vector $\boldsymbol{\nu}_m(T^{-1}(\hat{\boldsymbol{x}}_m - \boldsymbol{y}_m))$ with basis functions $\boldsymbol{g}_{m1}, \ldots, \boldsymbol{g}_{mM}$.
- If λ is close to zero, $\hat{\boldsymbol{v}}_m$ is estimated as the residual vector $T^{-1}(\hat{\boldsymbol{x}}_m) - \boldsymbol{y}_m$ without smoothing.

To compute the optimal $q_2^\star(\boldsymbol{c}, \boldsymbol{e})$, we fix $q_1(\boldsymbol{v}, \boldsymbol{\alpha})$ and $q_3(\rho, \sigma^2)$, put the full joint into the general solution of VBI as before and collect all terms containing $\boldsymbol{c}$ and $\boldsymbol{e}$ which arise in the two-component mixture and the fact that $c_n = \sum_{m=1}^{M} c_n \delta_m(e_n)$

$$\ln q_2^\star(\boldsymbol{c}, \boldsymbol{e}) = \sum_{n=1}^{N} \mathbb{E}_{q_1 q_3}[\ln p(\boldsymbol{x}_n, c_n, e_n | \boldsymbol{y}, \boldsymbol{v}, \boldsymbol{\alpha}, \rho, \sigma^2)] + \text{const.}$$

$$= \sum_{n=1}^{N} \mathbb{E}_{q_1 q_3}\left[\ln\left((w p_{out}(\boldsymbol{x}_n))^{1-c_n} \left((1-w) \prod_{m=1}^{M} (\alpha_m \phi_{mn})^{\delta_m(e_n)} \right)^{c_n} \right)\right] + \text{cons}$$

$$= \sum_{n=1}^{N} \left[\ln(w p_{out}(\boldsymbol{x}_n))^{1-c_n} + \sum_{m=1}^{M} c_n \delta_m(e_n) \ln((1-w)\mathbb{E}_{q_1 q_3}[\ln(\alpha_m \phi_{mn})]) \right] + \text{c}$$

$$= \sum_{n=1}^{N} \left[\ln(w p_{out}(\boldsymbol{x}_n))^{1-c_n} + \sum_{m=1}^{M} \ln((1-w)\langle\alpha_m\rangle\langle\phi_{mn}\rangle)^{c_n \delta_m(e_n)} \right] + \text{const.}$$

where $\langle\alpha_m\rangle = \exp(\mathbb{E}_{q_1 q_3}[\ln(\alpha_m)])$ and $\langle\phi_{mn}\rangle = \exp(\mathbb{E}_{q_1 q_3}[\ln(\phi_{mn})])$.
This implies that $q_2^\star(\boldsymbol{c}, \boldsymbol{e})$ is factorized into $q_2^\star(\boldsymbol{c}, \boldsymbol{e}) = \prod_{n=1}^{N} {q_2^\star}^{(n)}(c_n, e_n)$.

Using the linearity of the expected value and the law of logarithms we also have

$$\begin{aligned}\mathbb{E}_{q_1 q_3}[\ln(\alpha_m \phi_{mn})] &= \mathbb{E}_{q_1 q_3}[\ln \alpha_m + \ln \phi_{mn}] \\ &= \mathbb{E}_{q_1 q_3}[\ln \alpha_m] + \mathbb{E}_{q_1 q_3}[\ln \phi_{mn}]\end{aligned}$$

Applying $\ln(\exp(x+y)) = \ln(\exp(x)\exp(y))$ we have

$$\begin{aligned}\ln\exp(\mathbb{E}_{q_1 q_3}[\ln(\alpha_m \phi_{mn})]) &= \ln\exp(\mathbb{E}_{q_1 q_3}[\ln \alpha_m])\exp(\mathbb{E}_{q_1 q_3}[\ln \phi_{mn}]) \\ &= \ln(\langle\alpha_m\rangle\langle\phi_{mn}\rangle)\end{aligned}$$

which are required to update $q_2(\boldsymbol{c}, \boldsymbol{e})$.

Taking the exponential of $q_2^{\star(n)}(c_n, e_n)$ we can see that

$$q_2^{\star(n)}(c_n, e_n) \propto (wp_{out}(\boldsymbol{x}_n))^{1-c_n} \prod_{m=1}^{M} ((1-w)\langle\alpha_m\rangle\langle\phi_{mn}\rangle)^{c_n \delta_m(e_n)}$$

The normalization of $q_2^{\star(n)}(c_n, e_n)$ is obtained by the summation of the right-hand side for all pairs of $c_n \in [0, 1]$ and $e_n \in \{1, \ldots, M\}$ as follows:

$$wp_{out}(\boldsymbol{x}_n) + (1-w) \sum_{m=1}^{M} \langle\alpha_m\rangle\langle\phi_{mn}\rangle$$

The division of the r.h.s of $q_2^{\star(n)}(c_n, e_n)$ by the normalization constant is equivalent to the divisions of **all** component terms, i.e. $wp_{out}(\boldsymbol{x}_n)$ **and** $\{(1-w)\langle\alpha_m\rangle\langle\phi_{mn}\rangle\}_{m=1}^{M}$ by the normalization constant. This is because only one component term is active among the terms for each pair of c_n and e_n.

We define p_{mn} as the matching probability between $\boldsymbol{x}_n$ and $\boldsymbol{y}_m$ as

$$p_{mn} = \frac{(1-w)\langle\alpha_m\rangle\langle\phi_{mn}\rangle}{wp_{out}(\boldsymbol{x}_n) + (1-w)\sum_{m'=1}^{M}\langle\alpha_{m'}\rangle\langle\phi_{m'n}\rangle}$$

With that we have $\boldsymbol{\nu}'_n = \sum_{m=1}^{M} p_{mn}$ as the posterior probability that $\boldsymbol{x}_n$ is a non-outlier and $(1-\boldsymbol{\nu}'_n)$ that $\boldsymbol{x}_n$ is an outlier.

Therefore we obtain the closed form expression of $q_2^\star(\boldsymbol{v}, \boldsymbol{e})$ as a combination of a Bernoulli distribution and a categorical distribution as follows

$$q^\star(\boldsymbol{c}, \boldsymbol{e}) = \prod_{n=1}^{N}(1-\boldsymbol{\nu}'_n)^{1-c_n}\left(\boldsymbol{\nu}'_n \prod_{m=1}^{M}\left(\frac{p_{mn}}{\boldsymbol{\nu}'_n}\right)^{\delta_m(e_n)}\right)^{c_n}$$

where the posterior marginal distribution of c_n is the Bernoulli distribution with the probability $\boldsymbol{\nu}'_n$ and the posterior conditional distribution of e_n given $c_n = 1$ is the categorical distribution with probabilities $\left\{\frac{p_{mn}}{\boldsymbol{\nu}'_n}\right\}_{m=1}^{M}$

We have seen that we still need to compute the expectation $\langle \alpha_m \rangle$ and $\langle \phi_{mn} \rangle$ w.r.t $q_1(\boldsymbol{v}, \boldsymbol{\alpha}) q_3 = q_{\boldsymbol{v}}(\boldsymbol{v}) q_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}) q_3$.
Let

$$\mathrm{Dir}(\boldsymbol{x}|\boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{i=1}^{K} x_i^{\alpha_i - 1}$$

be a general Dirichlet distribution and $\alpha_0 = \sum_{i=1}^{M} \alpha_i$ then the marginal distributions of x_i are Beta distributed

$$x_i \sim \mathrm{Beta}(\alpha_i, \alpha_0 - \alpha_i) = \frac{1}{B(\alpha_i, \alpha_0 - \alpha_i)} x_i^{\alpha_i - 1} (1 - x_i)^{\alpha_0 - \alpha_i - 1}$$

where

$$B(\alpha_i, \alpha_0 - \alpha_i) = \frac{\Gamma(\alpha_i)\Gamma(\alpha_0 - \alpha_i)}{\Gamma(\alpha_0)}$$

is the **normalization constant** and $\Gamma(\cdot)$ is the gamma-function.

To compute the expected value $\mathbb{E}_{q_1 q_3}[\ln x_i]$ we have

$$\begin{aligned}
\mathbb{E}_{q_1 q_3}[\ln x_i] &= \int_0^1 \ln x_i \,\mathrm{Dir}(\boldsymbol{x}|\boldsymbol{\alpha}) dx_i \\
&= \int_0^1 \ln x_i \,\mathrm{Beta}(\alpha_i, \alpha_0 - \alpha_i) dx_i \\
&= \int_0^1 \ln x_i \frac{1}{B(\alpha_i, \alpha_0 - \alpha_i)} x_i^{\alpha_i - 1}(1 - x_i)^{\alpha_0 - \alpha_i - 1} dx_i \\
&\overset{\frac{d}{dx} a^x = a^x \ln a}{=} \frac{1}{B(\alpha_i, \alpha_0 - \alpha_i)} \int_0^1 \frac{d}{d\alpha_i} x_i^{\alpha_i - 1}(1 - x_i)^{\alpha_0 - \alpha_i - 1} dx_i \\
&= \frac{1}{B(\alpha_i, \alpha_0 - \alpha_i)} \frac{d}{d\alpha_i} \int_0^1 x_i^{\alpha_i - 1}(1 - x_i)^{\alpha_0 - \alpha_i - 1} dx_i \\
&= \frac{1}{B(\alpha_i, \alpha_0 - \alpha_i)} \frac{d}{d\alpha_i} B(\alpha_i, \alpha_0 - \alpha_i) \\
&= \frac{d}{d\alpha_i} \ln B(\alpha_i, \alpha_0 - \alpha_i)
\end{aligned}$$

$$\begin{aligned}
\frac{d}{d\alpha_i} \ln B(\alpha_i, \alpha_0 - \alpha_i) =& \frac{d}{d\alpha_i} \ln \frac{\Gamma(\alpha_i)\Gamma(\alpha_0 - \alpha_i)}{\Gamma(\alpha_0)} \\
=& \frac{d}{d\alpha_i} (\ln \Gamma(\alpha_i) + \ln \Gamma(\alpha_0 - \alpha_i) - \ln \Gamma(\alpha_0))
\end{aligned}$$

We consider $\alpha_0 - \alpha_i$ a constant when taking the derivative w.r.t. α_i, because $\alpha_0 - \alpha_i = \sum_{j \neq i} \alpha_j$ does not contain α_i anymore, but α_0 is not a constant in the last term.

$$\begin{aligned}
=& \frac{d}{d\alpha_i} (\ln \Gamma(\alpha_i) + \ln \Gamma(\alpha_0 - \alpha_i) - \ln \Gamma(\alpha_0)) \\
=& \frac{d}{d\alpha_i} \ln \Gamma(\alpha_i) + \frac{d}{d\alpha_i} \ln \Gamma(\alpha_0 - \alpha_i) - \frac{d}{d\alpha_i} \ln \Gamma(\alpha_0) \\
=& \frac{d}{d\alpha_i} \ln \Gamma(\alpha_i) - \frac{d}{d\alpha_i} \ln \Gamma(\alpha_0)
\end{aligned}$$

For the second term we simply apply the chain rule and get

$$\frac{d}{\alpha_i}\ln\Gamma(\alpha_0) = \frac{d}{d\Gamma}\ln(\Gamma(\alpha_0))\frac{d}{d\alpha_0}\Gamma(\alpha_0)\frac{d}{d\alpha_i}\alpha_0 = \frac{1}{\Gamma(\alpha_0)}\Gamma'(\alpha_0)\cdot 1 = \frac{d}{d\alpha_0}\ln\Gamma(\alpha_0)$$

we get for the derivative

$$\frac{d}{d\alpha_i}\ln\Gamma(\alpha_i) - \frac{d}{d\alpha_0}\ln\Gamma(\alpha_0) = \psi(\alpha_i) - \psi(\alpha_0)$$

where $\psi(\cdot)$ is the digamma function.

We thus compute the probability for the expectation over $\ln\alpha_m$ as

$$\langle\alpha_m\rangle = \exp(\psi(\kappa+\nu_m) + \psi(\kappa M + \hat{N}))$$

where $\hat{N} = \sum_{m=1}^{M}\nu_m$, which is the number of matching points between target and source point set.

We also compute $\langle \phi_{mn} \rangle$ with $\phi_{mn} = \frac{1}{\sqrt{\det\left(2\pi\sigma^2 \boldsymbol{I}_D\right)}} \exp\left(-\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2\right)$ and $\hat{\boldsymbol{y}}_m = T(\boldsymbol{y}_m + \hat{\boldsymbol{v}}_m)$ is the expectation of $T(\boldsymbol{y}_m + \boldsymbol{v}_m)$ over $q_1 q_3$ as

$$\begin{aligned}
\mathbb{E}_{q_1 q_3}[\ln \phi_{mn}] &= \mathbb{E}_{q_1 q_3}\left[-\ln\left(\sqrt{\det(2\pi\sigma^2 \boldsymbol{I}_D)}\right) - \left(\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{y}_m)\|^2\right)\right] \\
&= \mathbb{E}_{q_1 q_3}\left[-\frac{1}{2}\ln\left((2\pi\sigma^2)^D\right)\right] - \mathbb{E}_{q_1 q_3}\left[\frac{1}{2\sigma^2}\|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{y}_m)\|^2\right] \\
&= -\frac{D}{2}\mathbb{E}_{q_1 q_3}\left[\ln(2\pi\sigma^2)\right] - \frac{1}{2\sigma^2}\mathbb{E}_{q_1 q_3}\left[\|\boldsymbol{x}_n - \mathcal{T}(\boldsymbol{y}_m)\|^2\right] \\
&= -\frac{D}{2}\mathbb{E}_{q_1 q_3}[\ln(\sigma^2)] - \frac{1}{2\sigma^2}\mathbb{E}_{q_1 q_3}[\|\boldsymbol{x}_n - T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2]
\end{aligned}$$

Let us have a closer look at the second term, without $-\frac{1}{2\sigma^2}$

$$\begin{aligned}
&\mathbb{E}_{q_1 q_3}[\|\boldsymbol{x}_n - T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2] \\
&= \mathbb{E}_{q_1 q_3}[\|\boldsymbol{x}_n\|^2 - \boldsymbol{x}_n^T T(\boldsymbol{y}_m + \boldsymbol{v}_m) + \|T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2] \\
&= \mathbb{E}_{q_1 q_3}[\|\boldsymbol{x}_n\|^2] - \mathbb{E}_{q_1 q_3}[\boldsymbol{x}_n^T T(\boldsymbol{y}_m + \boldsymbol{v}_m)] + \mathbb{E}_{q_1 q_3}[\|T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2] \\
&= \|\boldsymbol{x}_m\|^2 - \boldsymbol{x}_n^T \hat{\boldsymbol{y}}_m + \mathbb{E}_{q_1 q_3}[\|T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2]
\end{aligned}$$

The last term can be expanded as follows

$$= \mathbb{E}_{q_1 q_3}[(s\boldsymbol{R}(\boldsymbol{y}_m + \boldsymbol{v}_m))^T (s\boldsymbol{R}(\boldsymbol{y}_m + \boldsymbol{v}_m))]$$
$$= \mathbb{E}_{q_1 q_3}[s^2 \boldsymbol{y}_m^T \boldsymbol{R}^T \boldsymbol{R}\boldsymbol{y}_m + 2s\boldsymbol{v}_m^T \boldsymbol{R}^T \boldsymbol{R}\boldsymbol{y}_m + s^2 \boldsymbol{v}_m^T \boldsymbol{R}^T \boldsymbol{R}\boldsymbol{v}_m + 2s\boldsymbol{t}^T \boldsymbol{R}(\boldsymbol{y}_m + \boldsymbol{v}_m) + \boldsymbol{t}^T \boldsymbol{t}]$$

All the linear terms in $\boldsymbol{v}_m$ get replaced by $\hat{\boldsymbol{v}}_m$ and only the squared term has to be considered next

$$= s^2 \boldsymbol{y}_m^T \boldsymbol{R}^T \boldsymbol{R}\boldsymbol{y}_m + 2s\hat{\boldsymbol{v}}_m^T \boldsymbol{R}^T \boldsymbol{R}\boldsymbol{y}_m + s^2 \mathbb{E}_{q_1 q_3}[\boldsymbol{v}_m^T \boldsymbol{v}_m] + 2s\boldsymbol{t}^T \boldsymbol{R}(\boldsymbol{y}_m + \hat{\boldsymbol{v}}_m) + \boldsymbol{t}^T \boldsymbol{t}$$

Using the trace "trick" from before we represent the scalar product

$$\boldsymbol{v}_m^T \boldsymbol{v}_m = \mathrm{tr}(\boldsymbol{v}_m^T \boldsymbol{v}_m) = \mathrm{tr}(\boldsymbol{v}_m \boldsymbol{v}_m^T)$$

And since the trace is a linear mapping, we can pull it outside of the expected value and continue using the definition of the covariance matrix $\mathrm{Cov}(\boldsymbol{v}, \boldsymbol{v}) = \mathbb{E}[\boldsymbol{v}\boldsymbol{v}^T] - \boldsymbol{\mu}_v \boldsymbol{\mu}_v^T$

$$\begin{aligned}
\mathbb{E}_{q_1 q_3}[\boldsymbol{v}_m^T \boldsymbol{v}_m] &= \mathrm{tr}(\mathbb{E}_{q_1 q_3}[\boldsymbol{v}_m \boldsymbol{v}_m^T]) \\
&= \mathrm{tr}(\mathrm{Cov}(\boldsymbol{v}_m, \boldsymbol{v}_m) + \hat{\boldsymbol{v}}_m \hat{\boldsymbol{v}}_m^T) \\
&= \mathrm{tr}(\mathrm{Cov}(\boldsymbol{v}_m, \boldsymbol{v}_m)) + \mathrm{tr}(\hat{\boldsymbol{v}}_m \hat{\boldsymbol{v}}_m^T) \\
&= \mathrm{tr}(\mathrm{Cov}(\boldsymbol{v}_m, \boldsymbol{v}_m)) + \hat{\boldsymbol{v}}_m^T \hat{\boldsymbol{v}}_m
\end{aligned}$$

We now have all the coefficients to formulate the expected value of the squared norm as

$$\mathbb{E}_{q_1 q_3}[\|\boldsymbol{x}_n - T(\boldsymbol{y}_m + \boldsymbol{v}_m)\|^2] = \|\boldsymbol{x}_n - \hat{\boldsymbol{y}}_m\|^2 + s^2 \operatorname{tr}(\operatorname{Cov}(\boldsymbol{v}_m, \boldsymbol{v}_m))$$

Finally $\langle \phi_{mn} \rangle$ is computed as

$$\langle \phi_{mn} \rangle = \phi(\boldsymbol{x}_m | \hat{\boldsymbol{y}}_m, \sigma^2 \boldsymbol{I}_D) \exp\left(-\frac{s^2}{\sigma^2} \operatorname{tr}(\sigma_m^2 \boldsymbol{I}_D) \right)$$

where σ_m^2 is the mth diagonal element of Σ.

To compute the optimal $q^\star(\rho, \sigma^2)$ given $q_1(\boldsymbol{v}, \boldsymbol{\alpha})$ and $q_2(\boldsymbol{c}, \boldsymbol{e})$ we can directly maximize the lower bound without using the general solution of VBI, due to the assumption that q_3 is a Dirac-delta function. It is characterized only by its mode.
The lower bound $L(q_3)$ is computed as

$$L(q) = \int q_3(\boldsymbol{\theta})\mathbb{E}_{q_1 q_2}[\ln p(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\theta})])d\boldsymbol{\theta} - \int q_3(\boldsymbol{\theta}) \ln q_3(\boldsymbol{\theta})d\boldsymbol{\theta} + \text{const}$$

assuming $q_3(\rho, \sigma^2) = \delta(\boldsymbol{\theta} - (\rho, \sigma^2))$ the second term is zero everywhere. The left part boils down to computing the expectation w.r.t. q_3 so we have

$$\begin{aligned} L(q) &= \int q_3(\boldsymbol{\theta})\mathbb{E}_{q_1 q_2}[\ln p(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\theta})]d\boldsymbol{\theta} \\ &= \mathbb{E}_{q_1 q_2}[\ln p(\boldsymbol{x}, \boldsymbol{y}, \rho, \sigma^2)] \end{aligned}$$

because it is characterized only by its mode.

We define $\boldsymbol{u}_m = \boldsymbol{y}_m + \boldsymbol{v}_m \in \mathbb{R}^D$ and take the expectation of the logarithm of the full joint distribution over $q_1 q_2$ to obtain the lower bound of the model evidence as follows:

$$\begin{aligned}
\mathbb{E}_{q_1 q_2}[\ln p(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{\theta})] =& \hat{N} \ln \det\left(2\pi\sigma^2 \boldsymbol{I}_D\right)^{-1/2} \\
& - \frac{1}{2\sigma^2} \sum_{n=1}^{N} \sum_{m=1}^{M} p_{mn} \mathbb{E}_{q_1 q_2}[\|\boldsymbol{x}_n - (s\boldsymbol{R}\boldsymbol{u}_m + \boldsymbol{t})\|^2] + \text{const.} \\
=& - \frac{\hat{N} D}{2}(\ln(2\pi) + \ln(\sigma^2)) \\
& - \frac{1}{2\sigma^2} \sum_{n=1}^{N} \sum_{m=1}^{M} p_{mn} \mathbb{E}_{q_1 q_2}[\|\boldsymbol{x}_n - (s\boldsymbol{R}\boldsymbol{u}_m + \boldsymbol{t})\|^2] + \text{const.}
\end{aligned}$$

Because q_3 is a Dirac delta function, a point mass that increases the lower bound can be found by the maximization of the lower bound w.r.t $\rho = (s, \boldsymbol{R}, \boldsymbol{t})$ and σ^2. Taking the derivative w.r.t σ^2 and equating to zero we obtain

$$0 = -\frac{\hat{N} D}{2} \frac{1}{\sigma^2} + \frac{1}{2\sigma^4} \sum_{n,m}^{N,M} p_{mn} \mathbb{E}_{q_1 q_2}[\|\boldsymbol{x}_n - (s\boldsymbol{R}\boldsymbol{u}_m + \boldsymbol{t})\|^2]$$

$$0 = -\frac{\hat{N}D}{2}\frac{1}{\sigma^2} + \frac{1}{2\sigma^4}\sum_{n,m}^{N,M} p_{mn}\mathbb{E}_{q_1 q_2}[\|\boldsymbol{x}_n - (s\boldsymbol{R}\boldsymbol{u}_m + \boldsymbol{t})\|^2]$$

$$= -\frac{\hat{N}D}{2}\sigma^2 + \frac{1}{2}\sum_{n,m}^{N,M} p_{mn}\mathbb{E}_{q_1 q_2}[\|\boldsymbol{x}_n - (s\boldsymbol{R}\boldsymbol{u}_m + \boldsymbol{t})\|^2]$$

$$= -\sigma^2 + \frac{1}{\hat{N}D}\sum_{n,m}^{N,M} p_{mn}\mathbb{E}_{q_1 q_2}[\|\boldsymbol{x}_n - (s\boldsymbol{R}\boldsymbol{u}_m + \boldsymbol{t})\|^2]$$

$$\Leftrightarrow \hat{\sigma}^2 = \frac{1}{\hat{N}D}\sum_{n,m}^{N,M} p_{mn}\mathbb{E}_{q_1 q_2}[\|\boldsymbol{x}_n - (s\boldsymbol{R}\boldsymbol{u}_m + \boldsymbol{t})\|^2] \coloneqq \mathcal{E}(\rho)$$

Minimization of $\mathcal{E}(\rho)$ maximizes the lower bound. Therefore, the maximization of the lower bound can be replaced with the minimization of $\mathcal{E}(\rho)$. Since $\boldsymbol{R}$ is a rotation matrix with $\boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}_D$, a point mass of interest is obtained by solving the constrained minimization problem

$$\hat{\rho} = \arg\min_{\rho} \mathcal{E}(\rho) \qquad \text{s.t. } \boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}_D$$

Minimizing $\mathcal{E}(\rho)$ is an instance of the problem described in[7]. It is very similar to optimal rotation search your may know from the ICP-algorithm. Let us introduce the notation regarding the mean-like vectors and covariance-like matrices to be convenient for describing the minimizer of $\mathbb{E}(\rho)$ as follows:

$$\bar{\boldsymbol{u}} = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m \hat{\boldsymbol{u}}_m, \qquad \boldsymbol{S}_{uu} = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m (\hat{\boldsymbol{u}}_m - \bar{\boldsymbol{u}})(\hat{\boldsymbol{u}}_m - \bar{\boldsymbol{u}})^T + \bar{\sigma}^2 \boldsymbol{I}_D$$

$$\bar{\boldsymbol{x}} = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m \hat{\boldsymbol{x}}_m, \qquad \boldsymbol{S}_{xu} = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m (\hat{\boldsymbol{x}}_m - \bar{\boldsymbol{x}})(\hat{\boldsymbol{u}}_m - \bar{\boldsymbol{u}})^T,$$

where $\hat{\boldsymbol{u}}_m = \mathbb{E}_{q_1 q_2}[\boldsymbol{u}_m] = \boldsymbol{y}_m + \hat{\boldsymbol{v}}_m$ and $\bar{\sigma}^2 = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m \sigma_m^2$

[7] A. Myronenko and X. Song. "On the closed-form solution of the rotation matrix arising in computer vision problems". In: *arXiv preprint arXiv:0904.1613* (2009).

Differentiating $\mathcal{E}(\rho)$ w.r.t t and equate to zero, we obtain the equation

$$t = \bar{\boldsymbol{x}} - s\boldsymbol{R}\bar{\boldsymbol{u}}$$

Substituting t into $\mathcal{E}(\rho)$, the objective function $\mathcal{E}(\rho)$ is rewritten as

$$\mathcal{E}(\rho) = \frac{1}{D}(s^2 \operatorname{tr} \boldsymbol{S}_{uu} - 2s \operatorname{tr} \boldsymbol{R}^T \boldsymbol{S}_{xu}) + \text{const}$$

Using the lemma in[8], the closed-form expression of the minimization-problem is derived as follows:

$$\begin{aligned}
\hat{\boldsymbol{R}} &= \Phi d(1, \ldots, 1, \det\left(\Phi\Psi^T\right))\Psi^T \\
\hat{s} &= \operatorname{tr} \hat{\boldsymbol{R}}^T \boldsymbol{S}_{xu} / \operatorname{tr} \boldsymbol{S}_{uu} \\
\hat{\boldsymbol{t}} &= \bar{\boldsymbol{x}} - \hat{s}\hat{\boldsymbol{R}}\bar{\boldsymbol{u}}
\end{aligned}$$

where Φ and Ψ are the orthogonal matrices obtained by **singular value decomposition** of $S_{xu} = \Phi S'_{xu}\Psi^T$ with the diagonal matrix of singular values, S'_{xu}.

[8] A. Myronenko and X. Song. "On the closed-form solution of the rotation matrix arising in computer vision problems". In: *arXiv preprint arXiv:0904.1613* (2009).

Denote the current deformed shape with the similarity transformation $\tilde{T}$ by $\hat{\boldsymbol{y}} = \tilde{T}(\boldsymbol{y} + \hat{\boldsymbol{v}})$ and the matrix of matching probabilities by $\boldsymbol{P} = (p_{mn})$. Substituting s, $\boldsymbol{t}$ and $\boldsymbol{R}$ with $\hat{s}, \hat{\boldsymbol{t}}$ and $\hat{\boldsymbol{R}}$ in $\mathcal{E}(\rho)$, we obtain the equation for updating σ^2 as follows:

$$\hat{\sigma}^2 = \frac{1}{\hat{N}D}(\boldsymbol{x}^T d(\tilde{\boldsymbol{\nu}}')\boldsymbol{x} - 2\boldsymbol{x}^T \tilde{\boldsymbol{P}}^T \hat{\boldsymbol{y}} + \hat{\boldsymbol{y}}^T d(\tilde{\boldsymbol{\nu}})\hat{\boldsymbol{y}}) + \hat{s}^2\bar{\sigma}^2$$

where $\tilde{\boldsymbol{P}} = \boldsymbol{P} \otimes \boldsymbol{I}_D \in \mathbb{R}^{DM \times DN}$, $\tilde{\boldsymbol{\nu}} = \boldsymbol{\nu} \otimes \boldsymbol{I}_D$ with $\boldsymbol{\nu} = \boldsymbol{P}\mathbf{1}_N \in \mathbb{R}^M$ and $\tilde{\boldsymbol{\nu}}' = \boldsymbol{\nu}' \otimes \mathbf{1}_D \in \mathbb{R}^{DN}$ with $\boldsymbol{\nu}' = \boldsymbol{P}^T\mathbf{1}_M \in \mathbb{R}^N$.

Data: $\boldsymbol{x} \in \mathbb{R}^{DN}$, $\boldsymbol{y} \in \mathbb{R}^{DM}$, $w, \lambda, \kappa, \gamma$
Result: $\hat{\boldsymbol{y}} = \tilde{T}(\boldsymbol{y} + \hat{\boldsymbol{v}}) = s(\boldsymbol{I}_M \otimes \boldsymbol{R})(\boldsymbol{y} + \hat{\boldsymbol{v}}) + (\mathbf{1}_M \otimes \boldsymbol{t})$
Initialization: $\hat{\boldsymbol{y}} = \boldsymbol{y}$, $\hat{\boldsymbol{v}} = 0$, $\Sigma = \boldsymbol{I}_M$, $s = 1$, $\boldsymbol{R} = \boldsymbol{I}_D$, $\boldsymbol{t} = 0$, $\langle \alpha_m \rangle = \frac{1}{M}$;
$\sigma^2 = \frac{\gamma}{DNM} \sum_{n=1}^{N} \sum_{m=1}^{M} \|\boldsymbol{x}_n - \boldsymbol{y}_m\|^2, \boldsymbol{G} = (g_{mm'})$ with $g_{mm'} = \mathcal{K}(\boldsymbol{y}_m, \boldsymbol{y}_{m'})$;
while *not converged* **do**

Update q_2: $\langle \phi_{mn} \rangle = \phi(\boldsymbol{x}_n | \hat{\boldsymbol{y}}_m, \sigma^2 \boldsymbol{I}_D) \exp\left(-\frac{s^2}{2\sigma^2} \operatorname{tr} \sigma_m^2 \boldsymbol{I}_D\right)$;

$p_{mn} = \frac{(1-w)\langle \alpha_m \rangle \langle \phi_{mn} \rangle}{w p_{out}(\boldsymbol{x}_n) + (1-w) \sum_{m'=1} \langle \alpha_{m'} \rangle \langle \phi_{m'n} \rangle}$;

$\boldsymbol{\nu} = \boldsymbol{P}\mathbf{1}_N; \boldsymbol{\nu}' = \boldsymbol{P}^T \mathbf{1}_M; \hat{N} = \boldsymbol{\nu}^T \mathbf{1}_M; \hat{\boldsymbol{x}} = d(\tilde{\boldsymbol{\nu}})^{-1} \tilde{\boldsymbol{P}} \boldsymbol{x}$;

Update q_1: $\boldsymbol{\Sigma}^{-1} = \lambda \boldsymbol{G}^{-1} + \frac{s^2}{\sigma^2} d(\boldsymbol{\nu}); \hat{\boldsymbol{v}} = \frac{s^2}{\sigma^2} \tilde{\boldsymbol{\Sigma}} d(\tilde{\boldsymbol{\nu}})(\tilde{T}^{-1}(\hat{\boldsymbol{x}} - \boldsymbol{y}))$;

$\hat{\boldsymbol{u}} = \boldsymbol{y} + \hat{\boldsymbol{v}}; \langle \alpha_m \rangle = \exp(\psi(\kappa + \boldsymbol{\nu}_m) - \psi(\kappa M + \hat{N}))$;

Update q_3: $\bar{\boldsymbol{x}} = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m \hat{\boldsymbol{x}}_m; \bar{\sigma}^2 = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m \sigma_m^2; \bar{\boldsymbol{u}} = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m \hat{\boldsymbol{u}}_m$;

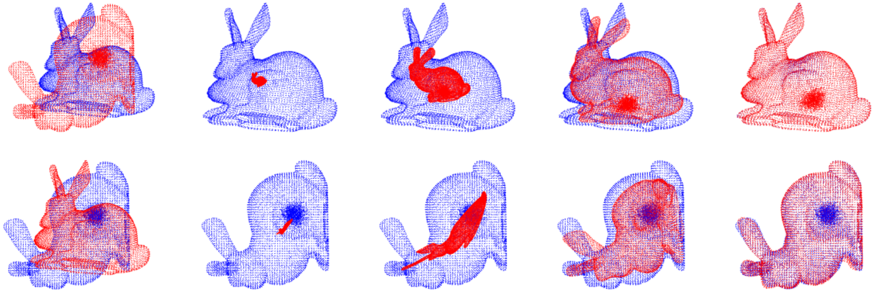
$\boldsymbol{S}_{xu} = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m (\hat{\boldsymbol{x}}_m - \bar{\boldsymbol{x}})(\hat{\boldsymbol{u}}_m - \bar{\boldsymbol{u}})^T$;

$\boldsymbol{S}_{uu} = \frac{1}{\hat{N}} \sum_{m=1}^{M} \boldsymbol{\nu}_m (\hat{\boldsymbol{u}}_m - \bar{\boldsymbol{u}})(\hat{\boldsymbol{u}}_m - \bar{\boldsymbol{u}})^T + \bar{\sigma}^2 \boldsymbol{I}_D$;

$\Phi \boldsymbol{S}'_{xu} \Psi^T = \operatorname{svd}(\boldsymbol{S}_{xu}); \boldsymbol{R} = \Phi d(1, \ldots, 1, \det(\Phi \Psi^T)) \Psi^T; s = \operatorname{tr}(\boldsymbol{R}\boldsymbol{S}_{xu}) / \operatorname{tr} \boldsymbol{S}_{uu}$;

$\boldsymbol{t} = \bar{\boldsymbol{x}} - s\boldsymbol{R}\bar{\boldsymbol{u}}; \hat{y} = \tilde{T}(\boldsymbol{y} + \hat{\boldsymbol{v}}); \hat{\sigma}^2 = \frac{1}{\hat{N}D}(\boldsymbol{x}^T d(\tilde{\boldsymbol{\nu}}')\boldsymbol{x} - 2\boldsymbol{x}^T \tilde{\boldsymbol{P}}^T \hat{\boldsymbol{y}} + \hat{\boldsymbol{y}}^T d(\tilde{\boldsymbol{\nu}})\hat{\boldsymbol{y}}) + s^2 \bar{\sigma}^2$;

end



Optimization trajectories for bunny data with rotation and clustered outliers. The points in the target and deformed shapes are colored blue and red, respectively. The leftmost column shows the initial point sets, and the optimization proceeds from left to right.

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